

# Chapter 1

## Introduction

### 1.1 Introduction

The fourth-third millennia BC cal (henceforth all dates BC cal will be indicated as BC) in the Near East saw the emergence of the worlds first large urban centres and states, well documented in Mesopotamia and Egypt <sup>1.1</sup>. Areas adjacent to Mesopotamia and Egypt, such as the Levant and Anatolia, witnessed significant contemporary changes but debate has developed over whether these changes represent another example of Mesopotamian style urbanism or an alternative process of social change (e.g. Falconer 1995, Fall et al. 1998, Finkelstein and Gophna 1993, Finkelstein 1995, Flannery 1972, Richard 1987). This thesis will directly address the question of Levantine urbanism in the late Fourth-early Third millennium BC by examining landuse and demography at the site of Tell esh Shuna. In doing so, I will also raise important questions regarding how fourth-third millennia communities engaged with, and impacted, their landscapes and how landscape transformations may have in part reflected the distinctive identities of new sorts of communities.

In particular, the Early Bronze Age (EBA) of the southern Levant, c. 3600 - 1900 BC, has long been regarded as an important transitional period of increasing social complexity (Joffe 1991, 3 and Philip 2001, 163) that witnessed the emergence of larger settlements, fortifications, and large public buildings. These developments have been conventionally interpreted as indicators of the first urban communities in the Southern Levant, and are important to this research because of the differing implications they present for land-use.

A number of authors have argued that these settled communities represent small-scale versions of the fourth and third millennium city-states of southern Mesopotamia and northern Syria (e.g. Ben-Tor 1992a). Ben-Tor's position implies that population densities at urban sites would out-strip local carrying capacities leading to interdependencies between sites and site hierarchies based on the exchange of food products for

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<sup>1.1</sup>Cowgill (2004, 526-528) provides an excellent discussion of many of the issues with the use of the terms 'urban' and 'urbanism'

'higher order' goods and services (e.g. crafts, trade items, military protection, judicial and religious services etc.). More recently, it has been suggested that the communities of the Southern Levant are no more than 'corporate villages', however, their development still suggests significant degrees of modification of the landscape by these communities and that for the first time communities 'inscribe' themselves on the landscape (Chesson 2000, Philip 2001). Philip and Chesson seem to suggest that demographic pressure (implied in Ben-Tor's model) never occurred in EBA Levant and that changes in settlement patterns and sites reflect changes at the 'ideological' level of organising communities and constituting and representing group identities. These propositions have neither been examined closely in the light of archaeological data concerning land use, nor have effective methodologies for the examination of such issues been employed. No one has actually tested in an empirical manner whether or not EB population densities would have put significant stress on the local carrying capacities given the production systems in use. This thesis aims to redress this.

It is argued that the EBA in the Levant witnessed widespread intensified production of what has become known as the classic triad of the Mediterranean crop economy (namely viticulture, olive and cereal production) (Fall et al. 2002, 451-454). The mass production of olive oil and wine in this period is supported by pottery finds (Stager 1985, 176-177), and by palynological studies, which have indicated a marked increase of olive production coinciding with decreasing natural forest cover (Baruch 1986). The nature of the interrelationships between settlement growth and distribution, and the appearance of these combined agricultural practices is an important issue (Esse 1991, Finkelstein and Gophna 1993) that will be explored in this thesis. This thesis aims to examine the significance of the growth of a network of large dense communities in terms of their exploitation of, engagement with, and impact upon their landscapes and to develop a new methodology for investigating these issues by creating models of likely sustaining areas in the landscape around sites and of the agricultural strategies and land use patterns employed by those communities. It is shown in this thesis that examining areas needed to sustain such communities and potential land use patterns required for the production of a settlements crops and animals allows for a more nuanced understanding of settlement growth, site distribution, interactions between communities and potential impacts on the environment, all of which pertain to questions of urbanism and impact.

Changes in subsistence strategies and settlement patterns are largely determined by people's ability to adapt and by their ability to manipulate environmental constraints, both of which can be linked to social, political and economic organisation (Esse 1991, 1). The research presented in this thesis furthers our understanding of the significance of developments from the 5th to the 3rd millennia by examining these changes and what may have prompted them. Using subsistence scenarios based on the archaeo-

logical record a computer program has been developed which calculates minimum and maximum catchment areas and maps relevant catchments most proximate to sites studied. Over one hundred sites in the region are modelled using these methods and the results incorporated into a path network/territory analysis. The systematic analysis of the likely sustaining areas of many broadly contemporary sites, together with their potential travel route network and approximate footprint sizes, results in maps which can indicate the degree of landscape exploitation, level of overall impact, possible areas where communities had to interact closely in relationship to access of areas of the landscape, and thus ultimately contribute to an analysis of the way in which people and communities were interacting in the southern Levant during the Chalcolithic and EBA periods. It thus goes some way toward answering questions about how these settlements were organised, both socially and economically, their status as urban centres, and the way in which they engaged with the landscape. In turn, this will provide new insights into whether EBA settlements were truly organised in a fundamentally different ('urban') way to those of the Chalcolithic. Section 2.8 on page 26 in Chapter 2 discusses the concept of urbanism in more detail.

## 1.2 Research Questions

As I have indicated the Early Bronze Age saw the appearance of a significant number and density of large settlements, to a degree not witnessed earlier anywhere in the world except Mesopotamia. In the light of the issues and approach I have outlined in Section 1.1, I have developed a number of specific questions, answers to which should provide insights into these phenomena.

It is proposed that agricultural intensification (Philip 2001) must have been associated with the development of these communities; is it possible to find evidence supporting this proposition? Did they outgrow the capacity of the immediately adjacent local landscape to support themselves?

Given the new larger scale of EBA communities, what sort of impact on the landscape can we understand such communities to have had? Are these likely to have been some of the first communities to have had a significant negative impact on their physical environment? Can we find evidence that, as Philip (2003) and Chesson (2003) put it, such communities 'inscribed' themselves on their landscape in new ways?

Much debate exists as to whether we should see such communities as 'urban'. If we understand urban communities as (at least partly) requiring access to resources to sustain them beyond those afforded by their immediate hinterland, then insight into the above question of whether these Early Bronze Age communities could be sustained by their immediately adjacent local landscape will surely be a valuable contribution to this debate. Can an empirically rigorous argument be made that, contrary to the

main trend in scholarly literature on the Early Bronze Age over the past 15 years, landpressures under conditions of population growth in the EBA favoured the development of regional site dependencies, exchange networks and perhaps also settlement hierarchies?

In addressing these issues, other questions arise. How did such relatively large communities sustain themselves within the environments in which they operated? What sorts of land use can we interpret from the available data on subsistence practices combined with environmental evidence and our understanding of their landscapes? If we can ascertain the types of land use, what scale of activity relating to the associated land use can we understand to have operated in these circumstances?

The use of site catchment analysis helps address several of these questions, but previous 'site catchment' studies have generally only focussed on individual settlements; however, settlements and communities exist in networks. In particular, one of the notable features of the Early Bronze settlement record is the obvious existence of networks of communities, especially in the larger communities (Philip 2001, 207). What can a combined study of land use and routes connecting a network of settlements allow us to understand about relationships between settlements and the nature of such networks?

The above questions generate important methodological and practical issues. In particular, an effective and efficient way to model multiple scenarios of land use from archaeological and landscape data to compensate for the imprecision inherent in the archaeological record has to be developed. Additionally, an effective way to model land use in reference to a network of settlements will be required.

### **1.3 Case Study**

It is rare to find a site in the Southern Levant that provides a good chronological record of continuous occupation spanning from the Chalcolithic to beyond the EB I. Tell esh Shuna North (henceforth simply Shuna) is one of these rare sites, with a sequence of archaeology from c. 3900-3000 BC cal. Regardless of how they explain settlement change, all scholars recognise EB I as a period of significant change in the Southern Levant (marked, for example by a growth in site size and the first appearance of fortified settlements). In addition to its unbroken chronology for the periods of interest to this thesis, Shuna was a large settlement (~ 20 hectares by the Late Early Bronze Age I) that was located at an important junction of major north-south and east-west trade routes (Philip and Baird 1993, 15) so an understanding of this settlement's subsistence strategies will enhance the understanding of its role in the regional network analysis carried out in Chapter 6.

Furthermore, excavations of the site conducted by Baird and Philip in the early 1990's provide detailed faunal and palaeobotanical records, crucial for the reconstruc-

tion of diet and land use, which is, as the reader will see, a critical component of my methodology. These factors make Shuna an ideal Chalcolithic - EBA settlement with which to explore the development of large 'urban' centres, to assess the scale of their impact on the environment, and the degree to which EBA settlements of the southern Levant were interacting both intra-regionally and inter-regionally.

## 1.4 Thesis Structure

The discussion and analysis in this thesis starts with the questions posed by research on the Early Bronze Age, then considers the methodology needed to answer such questions, develops a methodology original to this research, provides applications of the methodology in case studies of the EB I, and ends with reflections on the effectiveness of the methodology and provisional answers to questions posed in this chapter in the conclusions of Chapters 5, 6 and 7. More specifically this chapter is followed by Chapter 2, which provides a brief overview of the archaeology of the Early Bronze Age defining in greater detail the questions outlined above which provided the starting point for this research. A detailed discussion of the geographical setting and climate of the southern Levant is also presented, which provides a firm foundation upon which the subsequent computer modelling and landscape analysis work are based.

Every attempt is made in this thesis to guide the reader through the complicated computer-based methodology in a logical, step-wise, and sequential fashion. The research in this thesis draws heavily upon the use of Geographic Information Systems (GIS) and Chapter 3 begins by providing the reader with a broad overview of its use and application in archaeology, as well as definitions for much of the specific GIS terminology that is necessarily used throughout this thesis. A thorough explanation of the methodology applied to the EBA of the southern Levant is the focus of Chapter 4.

Chapter 4 introduces Landuse Analyst; the software application which was specifically written in order to tackle the research questions addressed by this thesis. Landuse Analyst automates much of the methodology and allows for a fast, efficient and systematic solution to an otherwise very complicated and time-intensive process. No extant commercially available or open source software allows the intricate levels of modelling that was required for this research. Consequently, a considerable component of this project was the development of open source software that provides an interactive, fully functioning, complex (but easy to use) application for modelling land use. Importantly, as will become evident, this software will be shown to be a valuable tool which can be employed beyond the specific archaeological research presented in this thesis. The Appendix of this thesis includes the complete source code for Landuse Analyst<sup>©</sup>. This has been included to illustrate the complexity of this software and to ensure that even if the online code hosting site ceases to exist, this open source soft-

ware will still be available to anyone wishing to use it. The complete set of files and resources necessary to compile this software is currently available for download online at <http://code.google.com/p/landuseanalyst>.

Having presented the landscape modelling process, the efficacy of Landuse Analyst is then demonstrated by applying it to the site of Tel esh Shuna North (hereafter Shuna) in Chapter 5. Thirty six different landuse scenarios ('set' combinations of various input factors used by Landuse Analyst) are presented for each of the three chronological periods examined; the Chalcolithic, EEBI and LEBI. The results are presented in groups which show the effects that various combinations of input factors have on the resultant landscape use catchment areas. Following the presentation of the catchment areas, four of these scenarios (which are shown to consistently produce maximum and minimum extents) are used to create catchments for an additional 125 EB I sites in the region in Chapter 6. These sites are then analysed in terms of the ways in which they may have been connected in a network of potential pathways. Finally, the results of the landscape analysis derived using Landuse analyst are overlaid and compared with results obtained using more traditional methods (e.g. XTENT modelling) and the effectiveness of the methodology then discussed.

This thesis concludes with a discussion of the future potential for both the general landscape use methodology employed in this thesis, and specifically for Landuse Analyst, in Chapter 7, along with a summary and conclusions of this research.

## Chapter 2

# Physiography and Archaeological Background

“The natural environment provides the physical underpinning of the cultural landscape.” – Wilkinson (2003, 15)

### 2.1 Introduction

In the introductory chapter, I stated that this thesis presents a method for identifying and examining landuse patterns, environmental impact, and settlement interaction. As the main thrust of this research is focussed on landscape, a discussion of the natural environment is imperative. The use of the term ‘landscape’ also requires qualification, as there are many different aspects of landscape<sup>2.1</sup> (e.g. environmental, ecological, or cultural landscapes). Unless otherwise stated, throughout this thesis the term landscape refers to the physical aspects of the environment (primarily distance, area and slope)<sup>2.2</sup>.

This chapter presents a physiographical overview followed by a brief review of Early Bronze Age archaeology pertinent to this research, with particular emphasis on subsistence economy and settlement patterns.

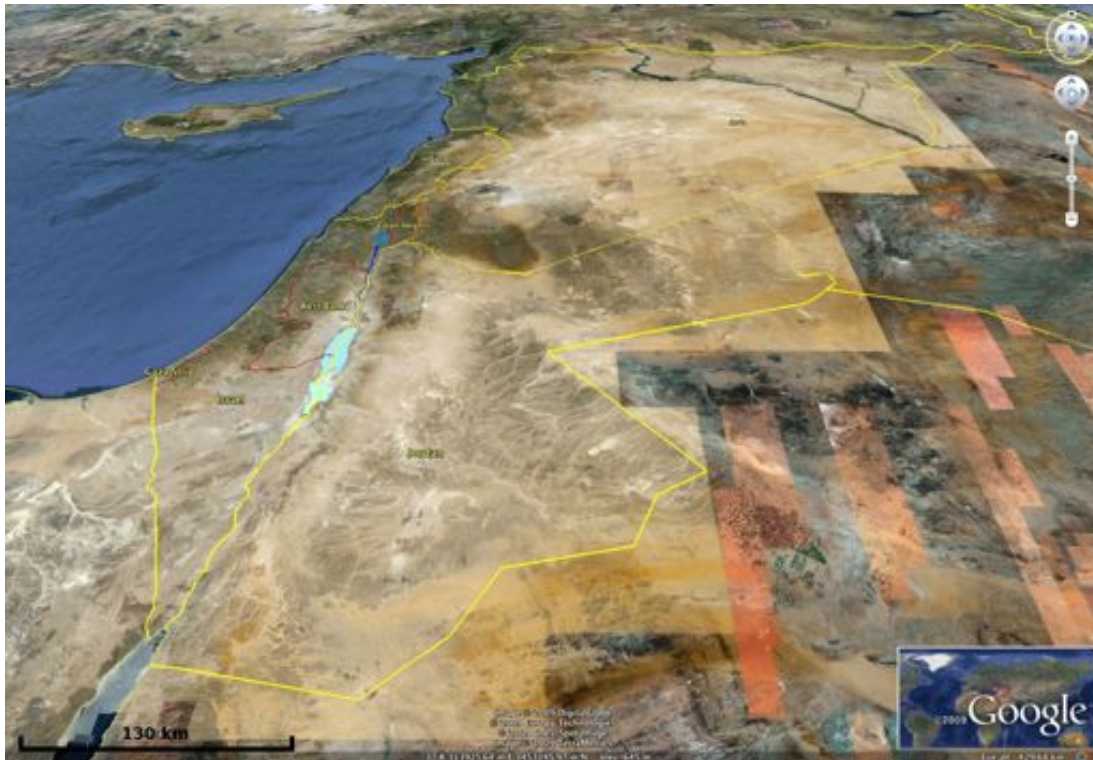
### 2.2 Physiography of the Southern Levant

Geographically, the southern Levant can be broadly defined as modern day Jordan, Palestine and Israel (see Figure 2.1). Although not a large region geographically ( 115,000 km<sup>2</sup>, ~ 500 km east/west by ~ 450 km north/south.), its location between Egypt and Mesopotamia guaranteed it a role in international trade, “...if not as an active participant then at least as a passive funnel through which merchants and armies

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<sup>2.1</sup>For an excellent discussion of the term landscape as used in archaeology as a discipline, see Anschuetz et al. (2001).

<sup>2.2</sup>The validity of physical geography and the physical landscape is emphasised by Wilkinson (2003, 15).



**Figure 2.1: The Southern Levant**

passed.” Esse (1991). Control of this region would have been enormously valuable and was surely an important part of the grand design of many empires Esse (1991).

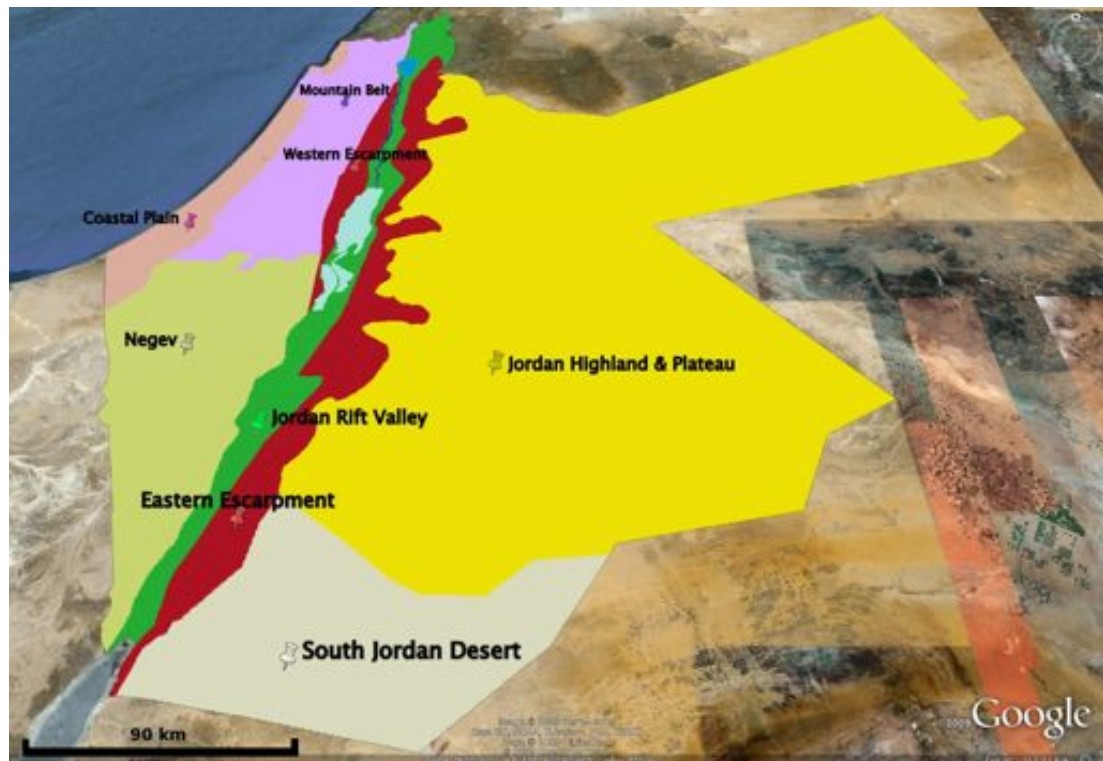
Figure 2.2 and the following descriptions of the seven physiographical regions outlined therein are all adapted from work conducted by the US Geological Survey for the Water Data Banks Project<sup>2,3</sup>.

**Coastal Plain** The Coastal Plain is located on the coast of the Mediterranean Sea. Formed by the emergent surface of the continental shelf, the Coastal Plain consists of thick Nile derived sediments covered by Quaternary eolian sands. It’s defining characteristics are a flat topography and a shoreline of white sand bordered by fertile farmlands (US Geological Survey 2009).

**Mountain Belt** This region is formed from layers of sedimentary rock. Originally deposited in horizontal layers, they were folded in southern and central areas whilst in the north, (including the mountains west of Lake Tiberias and their transverse valleys) they were offset by faulting. The Mountain Belt elevation ranges from 500 to 1,200 m above sea level (US Geological Survey 2009). As coastal winds rise over the

<sup>2,3</sup>Water Data Banks Project, Multilateral Working Group on Water Resources, Middle East Peace Process Overview of Middle East Water Resources <http://exact-me.org/overview/p0607.htm> (Accessed 10-May-2009)





**Figure 2.2:** Seven major physiographic regions of the Southern Levant, formed initially from tectonism and volcanism, evolving into the present form through the gradual processes of weathering, erosion and deposition (after US Geological Survey 2009).

mountains in the north, they cool, resulting in relatively high rainfall (see Figure 2.3).

**Negev** Northwest oriented mountains formed from Upper Cretaceous and Tertiary sedimentary rocks and several large east-west oriented faults characterise the northern Negev. The central Negev is characterised by low hills and plains, formed primarily by sandstone, making it an area susceptible to erosion, evident by numerous deeply incised wadis. These wadis typically flow only ephemerally today, often producing flash floods resulting from heavy winter rains. Volcanic craters, rocky plateaus, and rugged mountains dominate the landscape in the southern part of this region (US Geological Survey 2009). Generally, the entire Negev is a very arid region.

**Jordan Rift Valley** The rift valley, extending from Turkey to the Red Sea, plunges to 400 m below sea level at the modern day level of the Dead Sea, which is the lowest point on the surface of the earth. The valley, containing the Hula Valley, Lake Tiberias, and the Dead Sea, is the dominant physiographic and geologic feature of the Southern Levant. At 375 km in length and with a vertical displacement of more

than 3000 m, this strike-slip fault has a big affect on both the climate and the hydrology, subsequently impacting all anthropogenic activities in the region (US Geological Survey 2009). The valley north of the Dead Sea has a long history of agricultural use, almost certainly due (in part) to the abundance of water from both the Jordan River and the numerous springs along the edges of the valley.

**Western/Eastern Escarpments of the Jordan Rift Valley** As the Jordan Rift Valley deepened during its formation, abrupt valley walls and deep wadis formed across the escarpments, characterised by deep canyons incised through the Upper Cretaceous sedimentary rocks, and even reaching into the underlying Precambrian and Lower Cretaceous age rocks (US Geological Survey 2009).

**Jordan Highland and Plateau** The Jordan Highland consists mainly of deeply incised Cretaceous sedimentary rocks rising to 1,200 m in some places, and then dropping gradually towards the Jordan Plateau to the east at ~ 500 m above sea level. The Jordan Plateau is generally flat, open country, with shallow wadis draining inland to various depressions. Basalt flows have noticeably smoothed parts of this (US Geological Survey 2009).

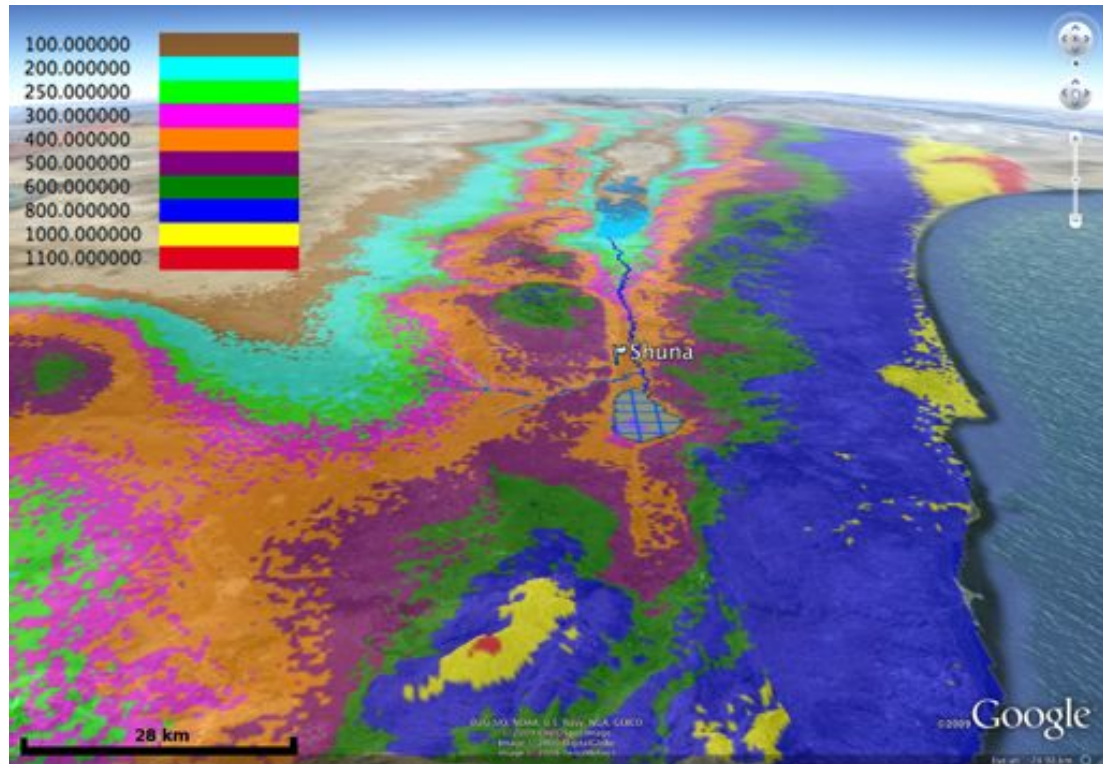
**South Jordan Desert** This extremely arid region is characterised by mountains formed from exposed Palaeozoic sandstone, dunes, and near the Dead Sea, exposed Precambrian crystalline rocks. Several prominent north-west/south-east oriented faults are also present here (US Geological Survey 2009).

The major bodies of water in the region are the Mediterranean to the West, the Red Sea in the South, Lake Tiberias in the North, the Dead Sea in the central region and the Jordan River, which flows from the southern end of Lake Tiberias southwards and drains into the Dead Sea.

## 2.3 Environmental Conditions and Changes

Many methods have been used for reconstructing the palaeo-climate of the southern Levant. Although much of the work has been focussed on Pleistocene climates, palaeoenvironmental data (derived from sources such as palynology, geomorphology, deep-sea cores, lacustrine deposits, speleothems, faunal evidence, and fossilised land snails Levy (1992, 68)) have begun to redress this imbalance (Rowan and Golden 2009, 20). Work done by Goldberg and Rosen (1987) indicates that the Early Holocene experienced two climatic phases, 9000-7000 B.P and again at around 5500 B.P. that were

characterised by wetter conditions than are typical of the southern Levant today. As Rowan and Golden (2009, 20) point out, however, although the data can tend to produce somewhat inconclusive interpretations it does indicate that conditions were generally more humid than today beginning in the Chalcolithic period. Characterised by



**Figure 2.3:** Modern annual rainfall in the Southern Levant (mm) (looking South)

a multitude of micro-environments and fluctuating seasonal rainfall patterns (Rowan and Golden 2009, 20), the southern Levant can be more generally divided into either a Mediterranean or Irano-Turanian climatic zone. Generally, the two zones can be classified with annual precipitation levels; the Mediterranean zone can be identified as areas which receive over 400 mm of rainfall per annum (natural woodland), and the Irano-Turanian zone (steppe) as areas receiving less than 400 mm (Bar-Yosef and Belfer-Cohen 1992).

Measurements of changes in the levels of prehistoric Lake Lisan (now the Dead Sea) have been used to determine overall shifts in the palaeoclimate, which has resulted in the general consensus that levels were higher between 4000 BC and 2500 BC than they are today (Rowan and Golden 2009, 20 and also page 108 of this thesis), implying greater moisture.

Although not a strong indicator of moisture levels, some inferences of climatic conditions can be made from examining subsistence strategies which are reflected in the archaeological record in the palaeobotanical and faunal assemblages (Rowan and Golden 2009, 21). Of the four primary domestic animal species (cattle, pigs, sheep and goats)

cattle and pigs require the most water, so when these animals are present in the archaeological record, it has been argued by some that it can indicate greater humidity (e.g. Grigson 1987, 1995, Hesse 1990). Crop preferences can also be used as indicators. For example, certain crops are more tolerant to drought (e.g. barley) and their increased presence can provide supporting evidence for a shift to a drier climate; this can also be inferred from an increase in other crops (e.g. legumes/vetches) which were possibly less desirable food sources, used as either animal fodder or emergency ‘famine food’ (Bourke 1997, 253). This supporting evidence is engaged with by Rowan and Golden (2009, 22), who suggest that the 250-300 mm rainfall isohyet of today lies in a similar position as it seems to have in the Chalcolithic.

However, much more reliable climatic proxy curves are available. Our understanding of environmental change has been much improved with the valuable multiproxy records that detail changes for many different indicators, including pollen, diatoms, microfossils, isotopes (both carbon and oxygen), and sedimentary mineralogy (Wilkinson 2003, 19). Using a combination of these and other factors, Wilkinson (2003, 22) summarises:

- there was a rapid increase in temperature and humidity both before and after the Younger Dryas
- the early to mid Holocene witnessed a moist interval
- after the moist period conditions became drier towards the end of the Bronze Age
- with considerable fluctuations, the last 4000 years have remained generally drier

Historical and sedimentological evidence from Egypt suggests that there was sustained drought in the late third millennium BC (e.g. Bell 1971). Further evidence exists which suggests that, during the same time period<sup>2,4</sup>, a sustained drought occurred in Syria (e.g. Courty and Weiss 1997, Weiss et al. 1993), indicating that this could have been a widespread, regional event (Fall et al. 2002, 461).

The distinct physiographical zones in the southern Levant allows a comparison of landscape engagement strategies between spatially proximate archaeological sites which differ in terms of their local physiography. Thus, differences in the culture and subsistence strategies observed at different sites can be assessed with less likelihood of differences being skewed from geographical distance/isolation.

It is apparent when looking at the zones that there is a strong correlation between the physiographical boundaries and patterns of precipitation; the latter being strongly linked to the topography of the region. This close relationship to the topography means that it is possible to look at rainfall patterns of the southern Levant today and apply

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<sup>2,4</sup>The late third millennium corresponds to the EBA IV.

the topographical effect back through time. For example, regions in modern day southern Levant having low and high amounts of precipitation would very likely have also had the same high/low relationship in the past given that the overall topography of the region has stayed the same (although it is likely that the boundary zones have shifted east/west somewhat). The (generally) predictable rainfall isohyet distribution thus makes it possible to generalise areas where lack of rainfall would have made dry farming only marginally possible, or conversely, where too much would have left water retained in the soil, resulting in swampy or saline land that is unsuitable for farming.

## 2.4 Subsistence Economy

“... questions of energy expenditures, or agricultural strategies, yield insights regarding the organization of society and production, as do studies of consumption and exchange. The traditional comparanda of cities, states, institutions and authority, technology, and religion, may also be discussed profitably at this level, along with gender, class, and ethnic dimensions.”

– Joffe (2004, 250)

From the Late Neolithic through the Early Bronze Age, subsistence economy patterns are very similar (Rowan and Golden 2009, 23). This is further supported by Bourke, who suggests that the conventional view of EBA innovations which would have boosted economic potential (such as niche fruit exploitation, agricultural intensification resulting from irrigation and ploughing, and long distance trading of staples facilitated by the use of equids) all can be seen to have been occurring prior to the Bronze Age (Bourke 2001, 116). Interestingly, this view seems to be in contrast with Philip who sees more innovation in EB I (Philip 2003).

If large settlements rely upon the surplus production of agricultural commodities provided by herders and farmers in the countryside, it necessitates a certain level of organisation between rural and urban people, and results in a hierarchical relationship (Fall et al. 1998). A hierarchical relationship develops as a result of time/labour division, role allocation, resource distribution, and the control/regulation of production (Fall et al. 1998). Subsistence practices are at the heart of this relationship, and a detailed examination of them can only help to improve our understanding.

“... despite periods of intermittent instability, sustainable agricultural systems have coevolved with humans in a mosaic of anthropogenic landscapes that have supported Near Eastern societies for thousands of years.”

– Fall et al. (2002, 447)

Agricultural specialisation can be seen in the archaeological record in a number ways, including crop or animal selection, agricultural practices or techniques, and tool

technology. Crop selection can indicate whether crops are being cultivated for subsistence use or for more commercial purposes.

Intensified farming, storage, redistribution, and craft specialisation are all indications of increasing economic complexity. The specialised production of pottery, metal, and lithics are all evidenced in the Southern Levant during the EBA. Craftsmen leave distinctive traces in the archaeological record, as they each had particular production techniques, technologies and tended to practice in different locations within the urban area. These artefacts are indicative of certain types of behaviour. For example, in the case of lithic specialists, it would be expected to see assemblages of flint or obsidian debitage, cores, raw materials, and/or knapping tools.

### 2.4.1 Crops

Cereal and legume cultivation was well established by the EBA. Rowan and Golden (2009, 23) list vital Chalcolithic crops as: hulled einkorn/emmer wheat (*Triticum monococcum* and *T. turgidum*), six and two row hulled barley (*Hordeum vulgare* and *H. distichum*), chickpeas (*Cicer arietinum*), lentil (*Lens culinaris*) and field peas (*Pisum sativum*). These species are seen to continue on in the EBA (e.g. EBA assemblages at Tel esh Shuna North: see Appendix A.1 on page 367)

The Old Testament states that Israel was blessed with seven main crop varieties, namely wheat, barley, grape, fig, pomegranate, olive and date (Deuteronomy 8:8). Of these seven crops, all but the wheat and barley are fruit tree crops, but their inclusion indicates the continued importance of wheat and barley into historical times, thereby reinforcing their importance to the people of the southern Levant in general.

Fruit is well suited to regional exchange, as it has relatively low bulk, is highly storable, and is easily transportable. These characteristics make fruit very important when looking at evidence in the archaeological record (Fall et al. 1998). Olives, grapes and figs are of particular importance, as they provided the resource for most horticultural products throughout the Mediterranean basin in areas which did not have access to irrigation but had to instead rely upon rain.

Olives were the most important fruit tree grown, as they provided oil which was used for many things in EBA life. Some of the uses for olive oil were cooking, use as an ingredient in ointments, and as a lamp fuel, evidence for which can be seen in the ceramic vessels. Combining the versatility of olive oil with the fact that it can be stored easily and for long periods of time made it a highly valued trade commodity (Fall et al. 1998) which in turn potentially drives another specialisation, that being olive oil production as opposed to the production of staple crops such as wheat or barley.

Grapes were another versatile orchard fruit. Unlike olives, grapes can be eaten fresh, but in this state they are poorly suited for storage or transport; the fruit must be processed for it to become a viable trade commodity. Grapes can be dried to make

raisins, or pressed to make juice or wine. Wine could be sealed in ceramic jars, stored, and in this state be easily transported making it ideal for regional exchange (Fall et al. 1998). Wine production can thus become another indicator for specialisation. This propensity for viticulture indicating specialisation is further reinforced with the suggested association of irrigation with olive and vine crops in the EBA (particularly in more arid regions, such as around the Dead Sea).

#### **2.4.2 Animal Husbandry**

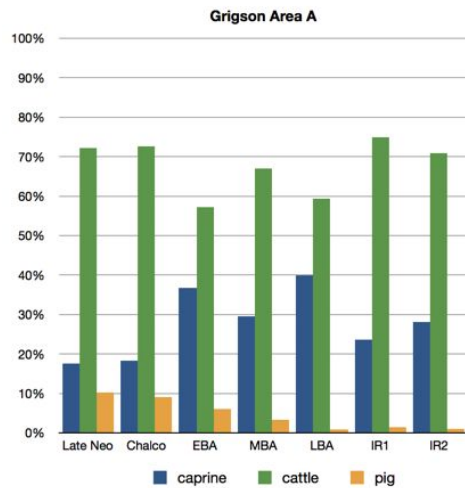
South Levantine pastoralism in the 4<sup>th</sup> and 3<sup>rd</sup> millennium is a subject which continues to be widely debated, particularly for the arid Negev desert region (e.g. Levy 1992, 69 and Rowan and Golden 2009, 25). During the fifth millennium BC, after millennia of sheep and goat herding, a symbiotic relationship formed between the now sedentary farming communities and specialised pastoral groups, a relationship that has persisted in some regions of the Middle East to the present day (Golden 2004, 46). Pastoral production witnessed a shift in focus with more emphasis placed on dairy goods, a development which has been called the “secondary products revolution” (Sherratt 1983, Levy 1992).

As will be discussed in Chapter 6, the suggestion of a relationship between farming communities and specialised pastoral groups may have been more complicated and reactionary than what exists today. For example, it will be proposed that these relationships may have formed from a migration of people from the main settlement to smaller new settlements (possibly only occupied part of the year) for convenience as pastures were pushed further and further away from their homes. It is also proposed in Chapter 6 that this could also have been linked to a desire for an increased human presence on increasingly valuable/distant grazing lands to secure the area as their own.

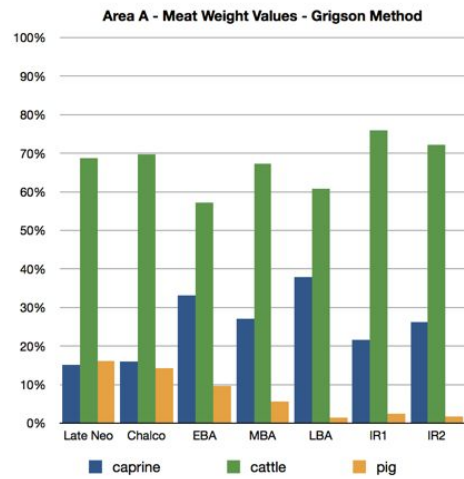
Animal husbandry can indicate specialised agricultural roles. Although goats or sheep are well suited as a commodity for exchange (they are both herd animals capable of secondary products such as wool and hair), their presence alone does not necessarily indicate anything in particular; the mere presence of pigs, however can. Pigs are difficult animals to herd or transport, and they provide no secondary products, making them poorly suited for exchange. Pigs are, however, very easy for single families to raise as they can more or less exist as domestic scavengers; for this reason, pigs can be diagnostic of sedentary household husbandry (Fall et al. 1998). The presence of pigs can also be used as “... a measure of a community’s disengagement from large-scale tributary or ‘market’ systems.” (Hesse 1990, Hesse and Wapnish 2001, 269).

Individuals accumulating large herds possibly constituted one of the first forms of ‘wealth’ in ancient times, which when accompanied by the control of grazing and crop land by particular individuals, can be perceived as the impetus for an emerging sense of territoriality and increased social complexity (Golden 2004, 46).

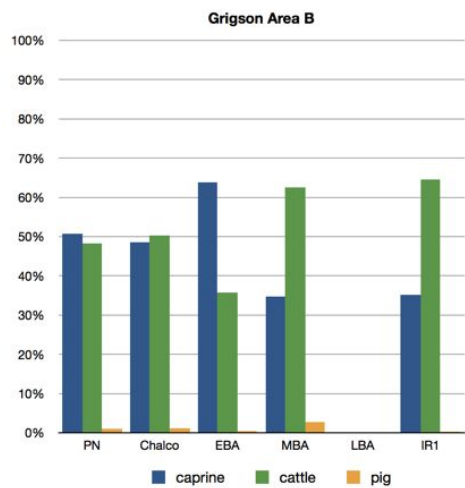




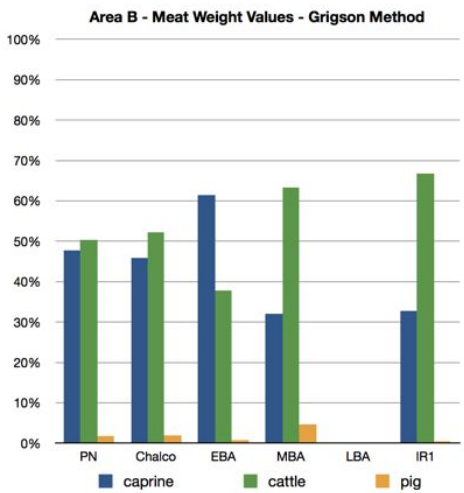
(a) Grigson Area A



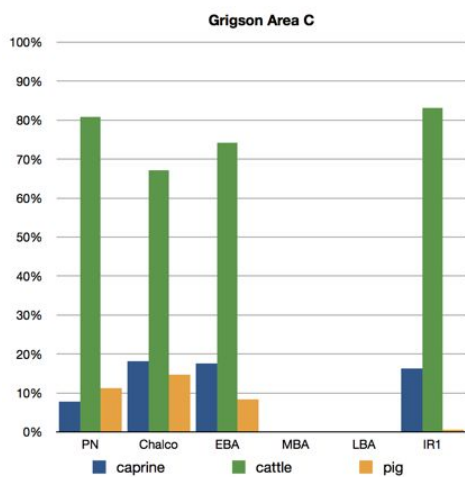
(b) Adjusted Area A



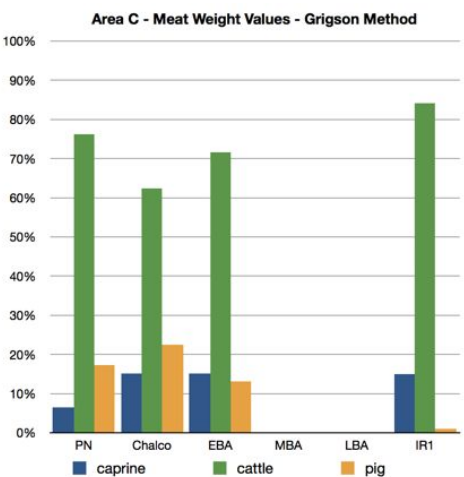
(c) Grigson Area B



(d) Adjusted Area B



(e) Grigson Area C



(f) Adjusted Area C

**Figure 2.4:** From Grigson (1998), relative animal contribution to the meat portion of the diet (a,c,e) which appear to have been errantly based on live animal weights. Figures (b,d,f) show the results of using the values she said she was intending to use. Note the large difference in pig contributions.



Grigson (1998, 251) states that although pigs did contribute up to 15% of the meat in the Chalcolithic at some sites, they were never of much importance as contributors to the diet. There is a serious error in her data, however, which noticeably changes the figures upon which she based her statement. Grigson (1998, 248, Figure 4) lists the meat weights of Sheep (80), Goat (80), Cattle (625), and Pigs (100) citing Clark and Yi (1983) as the source for these values. However, Table 2 from Clark and Yi (1983, 193, Table 2) was the source for these values (see Table 2.1). However, the latter source gives different values; it appears that she used the lowest value for live weight range of the animal instead of the usable weight, which is what she seems to have intended, and seemingly what it should be.

**Table 2.1:** Figures for animal weights after Clark and Yi (1983, 193)

| Species | Range kg | Mean Weight | % Usable | Weight Usable |
|---------|----------|-------------|----------|---------------|
| Sheep   | 80-110   | 95          | 50       | 47.5          |
| Goat    | 80-110   | 95          | 50       | 47.5          |
| Cattle  | 625-750  | 687         | 60       | 412.2         |
| Pigs    | 100-275  | 190         | 60       | 114.0         |

**Table 2.2:** Meat contribution from each animal, relative to cattle

| Species | Grigson                   | Clark                         |
|---------|---------------------------|-------------------------------|
| Sheep   | $\frac{80}{625} = 12.8\%$ | $\frac{47.5}{412.2} = 11.5\%$ |
| Goat    | $\frac{80}{625} = 12.8\%$ | $\frac{47.5}{412.2} = 11.5\%$ |
| Cattle  | $\frac{625}{625} = 100\%$ | $\frac{412.2}{412.2} = 100\%$ |
| Pigs    | $\frac{100}{625} = 16\%$  | $\frac{114}{412.2} = 27.7\%$  |

The significance of this error can be seen when a comparison of the animal meat contribution ratios are compared. To do this, it is easiest to normalise the values to cattle. These calculations can be seen compared in Table 2.2. Clearly, the pigs have been massively under-represented, and the caprines somewhat over-represented with respect to relative meat contributions. Using the actual values from the original Clark and Yi (1983) paper results in meat contributions from pigs nearly doubling from 16% to 27.7% (an increase of 42.2%).

It is important to point out this issue because, as discussed above, pigs can be used as indicators for many things, including subsistence strategies, and their importance as contributors to meat must not be understated. Pigs do not contribute dairy products and only provide meat. This suggests that when pigs are present, more reliance could be on meat. Pigs also have large litters and grow rapidly, meaning that breeding herd sizes can be smaller than animals which only have one or two offspring at a time. If

pigs are cut from the diet, and another species is used to ‘fill the void’, it would likely be cattle, goat or sheep, all of which have the ability to contribute dairy and because the breeding herd has to be larger to produce the same amount of meat with caprines, the *potential* for dairy products increases even more. This increased dairy potential is shown to have an affect on subsistence and land use strategies in Chapter 5.

## 2.5 Chronology

### 2.5.1 Terminology

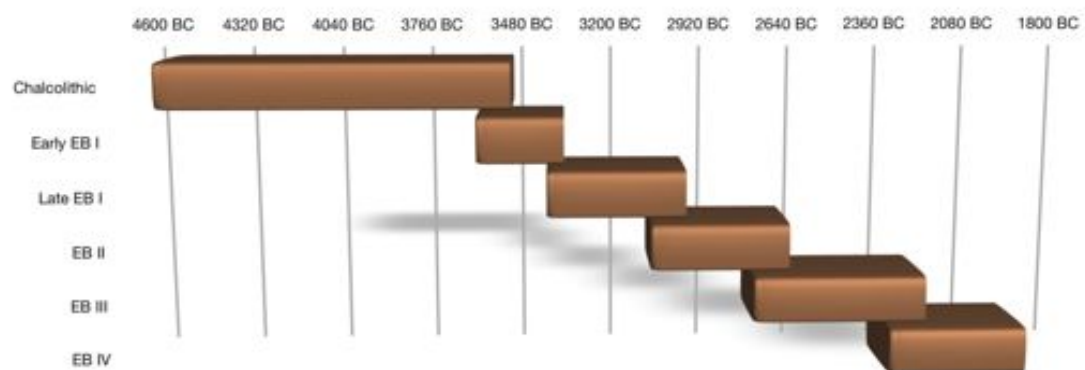
The terminology used when discussing the chronology of this area is based on the dominant material used for making tools. Thus, terms such as ‘Bronze Age’ have come to represent certain dates in an attempt to refine the framework of cultural developmental stages. The Chalcolithic period retained much of the material culture characteristic of the preceding Neolithic period, but due to the local inception of copper technology it has been given the name Chalcolithic (Gk. *khalkos* ‘copper’ + *lithos* ‘stone’). The Bronze Age, so named for the bronze technologies of the period, although tin bronze only appears in the southern Levant in EB IV, is further broken down into more specific periods: Early, Middle and Late. The Early Bronze Age of the Southern Levant can further be broken down into four more specific periods, represented with Roman Numerals. Naming conventions are such that these periods can easily be represented as ‘EB II’ for ‘Early Bronze Two’ or ‘EB IV’ for ‘Early Bronze Four’.

Largely due to marked regional difference in pottery, the terminology and definition of the EB I has produced a multitude of proposals, spawning much scholarly dispute (Philip 2001, 170) over which conventions should be used. Kenyon, of course, initially suggested the term Proto-urban (Kenyon 1959), Stager (1992) used a two-phase terminology (EB IA and EB IB) which was widely adopted for some time, but in the absence of radio carbon dates, it proved difficult to correlate the contemporaneity of materials dated as EB IA or IB between sites (Philip 2001, 170). Additionally, scholars have identified up to four phases in the EB I (e.g. Amiran and Gophna 1992, Table 1, and Levy 1995). Philip suggests that “Until reliable correlations have been established between regional pottery sequences, it seems better to simply allocate material to an earlier or later position within its own regional sequence.” (Philip 2001, 170). This suggestion has been used throughout this thesis (EEBI for Early EB I and LEBI for Late EB I).

### 2.5.2 Dates

The chronology of the Southern Levant in the Early Bronze Age has been derived from many different dating methods. It is important to recognise that relative dating alone can not determine when an event occurred, but only the order in which events have

occurred. For example, it can be argued that relative dating methods are generally only able to tell us how old something is if they can be connected to dates which have been derived from historical records or absolute methods<sup>2.5</sup>.



**Figure 2.5:** Dates used throughout this thesis when referring to the various periods.

An historical record of the sequence of dynasties in Egypt exists from the end of the 4th millennium BCE, and because artefacts of Egyptian origin are found on many sites in the Southern Levant, these artefacts can be used to establish dates (Philip 2001). As Philip points out, however, historical dating based upon Egyptian records has only been able to provide good evidence for the end of the EB I, before which the certainty of the dates for the Egyptian Dynasties lessens, thus becoming less reliable for establishing chronologies. For dates prior to this, therefore, radio carbon dates are commonly used.

Due to the different methods of establishing chronologies, there has been ranging proposals given regarding start and end dates for periods, however a rigorous sampling policy for radiocarbon analysis from excavations at Tell esh Shuna North in the Jordan Valley by Baird and Philip in the 1990's (see Section 5.3 on page 113) have resulted in a much more reliable chronology, as the site has been shown to have been continuously inhabited throughout these periods.

### **Radiocarbon Dates from Tell esh Shuna North**

At the outset of Baird and Philip's excavations at Tell esh Shuna North, the intent was to use radiocarbon analysis to answer a multitude of questions, including start and end dates of any periods represented during the excavation (Millard, forthcoming).

The results of this work indicate a start date for Chalcolithic activity of 5450-4930BC with the Early Chalcolithic period lasting until 4970-3900BC (Millard, forth-

<sup>2.5</sup>One example of this connection between absolute and relative dating techniques can be seen when examining the way in which dates can be derived from ceramic assemblages. An absolute method like radiocarbon dating can be used to quite precisely date organic materials enabling date ranges to be assigned to ceramic typologies when diagnostic pottery sherds are found contemporaneously with any organic materials. This explains how, even though a particular sherd may not be directly associated with other dateable material, a chronology can be determined from relative methods.

coming). Millard, forthcoming also puts the start of the EB I at 3750-3400BC. These dates for the Chalcolithic to EB I transition are consistent with (Millard, forthcoming) the generally accepted date of ~3500BC (Joffe and Dessel 1995, Stager 1992) and the 5000-4500BC suggested for the Ghassulian Early Chalcolithic (Joffe and Dessel 1995).

The evidence from Shuna also provides clear dates for the end of the Early EB I of 3620-3330BC (Millard, forthcoming). Due to a lack of suitable samples, a 330 year range puts the start of the Late EB I at either 3500-3310BC or 3290-3120BC (Millard, forthcoming). The Late EB I occupation lasts from 0-470 years and ends at 3350-3010BC (Millard, forthcoming). These dates for the EB I are, like the previous period, compatible (Millard, forthcoming) with the (3200-)3100BC for the EBA I to EBA II transition proposed by Joffe (1993). These dates, used throughout this thesis when referring to the various periods, are shown in Figure 2.5.

Having looked at the general chronology, a brief overview of some of the key developments and issues for the Chalcolithic and Early Bronze periods will now be presented.

## 2.6 Chalcolithic

“[The] Chalcolithic period in the southern Levant was a period of relatively rapid social change. During this time, we see the earliest evidence for a division of Labour, with at least some level of co-ordination for differentiated economic activities, and the advent of new technologies that impacted society in significant ways.”

– Golden (2004, 47)

The Chalcolithic period witnessed the emergence of many new sites, likely with several sub-regional culture variants (Golden 2004, 45). Golden (2004, 45) rightly points out that this ~ 1200 year period needs further chronological work, as many significant socio-economic transformations occurred during this time. Settlements appeared that were larger and more settled than seen previously (Golden 2004, 46). Rectilinear mud-brick and stone structures typify domestic architecture, but several sites (e.g. Gilat and Shiqmim) show evidence for structures which could have been used for housing extended families, or have been for use as public buildings (Golden 2004, 46). Chalcolithic sites range from smaller, apparently autonomous sites to large, 10+ ha sites (Rowan and Golden 2009, 27). The Chalcolithic gave rise to a large expansion of the number of sites, particularly in the more arid regions, suggested as being indicative of an increasing population more intensively exploiting new areas (Rowan and Golden 2009, 27 and Levy and Alon 1987). The Beersheva-Besor drainage basin illustrates this trend: only 11 Neolithic sites have been identified there, in contrast to 75 from the Chalcolithic Rowan and Golden (2009, 27).

The Beersheva valley and Jordan Valley region in general provide examples of large

Chalcolithic sites (e.g. Ghassul, Abu Matarand, Shiqmim and Bir es-Safadi), but the relationships between these large sites and the smaller ones is still being examined (Rowan and Golden 2009, 27). Surveys and limited investigations indicate that:

“... numerous large Chalcolithic sites such as Abu Hamid, Tall esh-Shuna, Tall Fendi, Pella, Tall Abu Habil, Tall al-Handaquq, and Kataret es-Samra were established along the valleys and the wadis feeding into the Jordan.”

– Rowan and Golden (2009, 27-28)

Bourke (2001, 113) suggests that there is a correlation between the size of the wadi flood-plain and the size of Chalcolithic settlements, with large sites being located on the larger wadi flood-plains while the smaller sites tend to be situated further up the wadi systems. Bourke (2001, 113) also notes that more than 90% of the large number of sites in the region are less than two hectares in size, and that the further east into the uplands one goes, both the size and frequency of sites diminish. Rowan and Golden (2009, 27) observes that other regions do not generally exhibit this degree of size differentiation.

Beth Shan, Gezer, Megiddo, and Tell Farah North from the central region have known Chalcolithic occupations, but there is limited data available due primarily to the deeply stratified nature of these tells, a result of long post-Chalcolithic occupations (Rowan and Golden 2009, 27). To the north, in the mountainous region in Upper and Lower Galilee, small sites (~ 0.5 ha) are most common, with only a few reaching one hectare in size, and even fewer reaching 4-5 ha (Rowan and Golden 2009, 28). The highlands provide no evidence for hierarchical settlements or large central sites (Gophna and Tsuk 2005); in the central highlands, sites are small and sparsely distributed in areas of higher elevation (Finkelstein et al. 1997) and likewise in the western foothills of the highlands in the west (Rowan and Golden 2009, 28). As mentioned above, the findings of Bourke (2001) in the uplands to the east make it unsurprising to find that this pattern of small, sparsely distributed settlements continues in this region with some exceptions (e.g. Sal). Archaeological work has tended to focus on the larger sites found in low lying regions, but Rowan and Golden (2009, 28) notes that there are exceptions: al-Khawarij, Abu Snesleh, Wadi Ziqlab and Sahab. For sites in the highlands (both east and west) there is little indication for any significant level of integration with autonomous site development (Rowan and Golden 2009, 29), and this trajectory continues into the EBA (Harrison and Savage 2003).

## **2.7 Early Bronze Age**

### **2.7.1 EB I**

There is a degree of continuity in the transition from the Chalcolithic to the EBA with the persistence of some stylistic and technological features (Golden 2004, 48). The

EB I in particular is viewed as the transitional period from a highly developed village level Chalcolithic to the larger settlements, fortifications, and large public buildings characteristic of the EB II. Together with the emergence of a market system based on specialised agricultural surpluses, it is the large, often fortified settlements of this period that are usually considered as the most notable developments of the Early Bronze Age (Golden 2004, 49).

Distinctive developments are evident in the EB I, none perhaps *dramatically* different from the preceding Chalcolithic, but coming together to suggest new social configurations. For example, large fortified sites (e.g. Ereni and Jawa) and monumental public architecture<sup>2.6</sup>. These large public buildings and fortified sites are characteristic of the EB II, yet are seen in the EB I, illustrating the transitional/transformative nature of this period.

Do these represent the presence of corporate institutions, that could have engaged with the landscape in distinctive ways alongside large scale houses or households? In this sense there is little to distinguish EB I and EB II/III, although fortifications and public buildings are more common in the latter. Further supporting the transitional nature of the EB I are changes in burial practices. For example, the large rock cut communal shaft tombs seen at EB I Bâb edh-Dhrâ and Jericho (Richard 1987, 25) are quite different from the Chalcolithic, but similar to those seen as more characteristic of the EB II. The mortuary landscape may have been becoming an increasingly important part of the way in which communities signalled their presence in the landscape, evident with use of, for example, dolmens (e.g. along the flanks of the Jordan valley) and nawamis (in Sinai) (Levy 1983).

Further, variability in households is suggested. This is not just by the range of *architectural types* (e.g. round and square buildings) but also by the range of *scale* of households. Thus, multiroom households with significant storage capacity at Erani and Beth Sean contrast with the single room broadroom structures common at most sites.

How effective are concepts of communities inscribing themselves on the landscape? Can we suggest they made such statements of ‘corporate grandeur’?. Or is it a matter of *incidental* impacts with agricultural intensification simply the by-product of population growth? What demands did new scales of communities make on the landscape? Can we think of intensification rather than changes in extent of land use and what are the implications of the subsistence practices we can detect for the nature of land exploitation? Can we understand how this is related to the mortuary landscape that may develop at this time?

There has been considerable debate over the nature of the EB I period in the southern Levant which we now consider to run from 3600-3000 BC cal. Thus some suggest

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<sup>2.6</sup>Public buildings are best attested with the so called ‘temples’ of EB I Megiddo, that required large scale construction effort and certainly operated at a monumental scale.

considerable continuity with preceding Chalcolithic (Hanbury-Tenison 1986), reflected in the way in which this period has even been referred to as Late Chalcolithic. Others saw more profound changes even from early in the history of archaeological research, hence Kenyon's proto-urban designation. Changes in architecture, namely curvilinear buildings, and new forms of pottery, Grey Burnished ware, were seen as indication of arrival of incomers, classically northerners (Kenyon 1959).

Other changes are considered significant. There appear major developments in the scale of metallurgy from early 4th millennium (Genz 2000). Sites in the southern Arabah produce copper for Egypt (e.g. Hujeirat al Ghuzlan). Major smelting sites appear for the first time near sources (e.g. Fidan 4). Secondary production centres like Shuna and Afridar appear and some, like Shuna, tap into northern metal sources in addition to the southern ones. The range of metal goods increases; weapons appear, and axe/adzes for agriculture likely became much more common, a view which is supported by the disappearance of their flint and ground stone counterparts from the archaeological record. Such 'metal-provisioned' agriculture suggests new economic relationships underpinning it.

Exchange seems to proliferate in EB I, with widespread evidence of primary and secondary metal production centres and exchange of copper implied. Silver crops up in the southern Levant (presumably from Anatolia). Many EB I vessels appear in Egypt. Many of these are storage/transport vessels and could have contained oil, wine and other agricultural products with significant implications for agricultural production strategies.

Nile valley produced vessels are also found in south Levantine sites. Tabular scrapers are mined and produced in the southern Jordanian deserts and imported to areas further west in significant quantities (Masashi 2008, Quintero et al. 2002, 327). Canaanite blades seem to be produced at regional centres and distributed from there. Some pottery probably circulates including grey Burnished wares and Line Painted wares.

### **2.7.2 EB II/III**

The EB II has been conventionally viewed as witnessing the appearance of the first 'urban' communities in the Southern Levant, although the use of the term 'urban' is a point of debate amongst scholars. Ben-Tor (1992a) for example argues that these settled communities represent small-scale versions of the third millennium city-states in southern Mesopotamia and northern Syria. More recently, (Chesson 2000 and Philip 2001) have suggested that the communities of the Southern Levant are no more than 'corporate villages' (further discussion of this can be found on page 25 of this thesis).

Many of the sites established in the EBA which were repeatedly occupied in the ensuing millennia became important tell sites, a phenomenon which Golden (2004, 48)

attributes to exceptionally well suited physical environments at these locations. This trend, however, results in difficulties for archaeologists, in that their EBA layers tend to be basal which makes it difficult to perform broad horizontal excavations (Golden 2004, 48).

Along with a sharp population increase, an overall shift in settlement to areas with more fertile land occurred in the EBA (Golden 2004, 48). Sometimes the start of the EBA is regarded as a period of cultural decline but by the end of fourth millennium significant socio-economic change had occurred (Esse 1989, Kempinski 1992). Joffe (1991, 4) suggests that the pattern of developing and collapsing social complexity that persisted in the southern Levant for three millennia had its starting point in the EB I. Subsistence strategies where the bulk of production takes place within the domestic family (introduced as 'domestic mode of production' by Sahlins 1972) formed the essential building block of EBA society (Joffe 1991, 6) and for both the rise and collapse witnessed in this period is the most useful concept for understanding EBA south Levantine social organisation (Joffe 1991, 6).

Factors affecting change in culture include technology, economics, political systems, ideologies and religion (Steward 1955). Golden (2004, 48-49) attributes most of the economic development in the EBA to significant advances in agricultural technologies which allowed for the production of staple food surpluses thus allowing the development of specialised craft production in such areas as metallurgy and ceramics. In this regard, Joffe reinforces what Golden says, but introduces international trade as an additional factor:

"A new pattern of settlement and series of technological and agricultural-economic factors, as well as international trade, also had roles in focussing society towards increased complexity."

– Joffe (1991, 4)

Surplus production of horticultural crops allowed for trading of such processed foods as wine and olive oil, and served as the catalyst for a lively trading relationship with Egypt (Golden 2004, 49). Joffe (1991, 4) suggests that, especially in the south, the EB I arose in association with the much more substantial Egyptian polity and that this association was an important factor in the course of rising social complexity. A sustained Egyptian presence is indicated by the large amounts of Egyptian pottery, seal impressions and architecture at several south Levantine sites and is reinforced with the presence of south Levantine pottery forms (i.e. Line Group Painted Wares, and wavy ledge-handled pots) at many sites in Egypt (Golden 2004, 49). This Egyptian connection provides historical chronological evidence beginning at the end of EB I and gives a firm correlation to the beginning of Egyptian Dynasty 1 (Philip 2001, 168).

In the north, 'Grey Burnished Ware' and 'Impressed-Slash Wares' are diagnostic of EEBI whilst the LEBI is characterised by pottery which displays 'Band-Slip' or 'Grain-



Wash' decoration (Philip 2001, 170). In the south, the EEBI is distinguished by the use of white washed vessels and red paint that appears to develop into the 'Line Painted Group' (sometimes called Line Group Painted Ware or LGPW) (Philip 2001, 170). LGPW is characteristic of the central regions during the LEBI (Joffe 1993). Philip (2001, 170) points out that while LGPW is well known in LEBI south-central Jordan, a distinct regional group is evident in EEBI Bab adh-Dhra'. After the EB I there is a decline in this ceramic regionalism and throughout the entire southern Levant, pottery generally became increasingly homogenous (Philip 2001, 170).

Philip (2001, 170) raises several issues with pottery chronologies in the EBA, but states that if one assumes that the chronologies are even broadly correct, settlements were often only occupied for part of the EBI-III. He continues on, suggesting that these occupations are assumed in the literature to span the entire period to which they are assigned, when in fact the mere presence of material at a site does not mean continuous occupation, as the material can rarely be placed more precisely than within a range of several hundred years (Philip 2001, 170); given this assessment, he suggests that conceptually, long term regional socio-economic relationships were unlikely to have been dictated by any one individual site. Philip concludes his discussion by suggesting that the more fluid settlement and political universe implied by this concept "... can be better related to a flexible, heterarchical world of corporate villages and middle-range societies than to the institutionalised power structures generally associated with city states." Philip (2001, 170).

Following the EB III, there was a region-wide collapse, which is often linked to socio-economic and political changes (e.g. Golden 2004, 51). The EB IV shows a clearly distinct culture from both the EB II-III and the Middle Bronze Age (Golden 2004, 51). Settlement patterns in the EB IV generally shifted to more rural regions as the larger centres were "... abruptly abandoned or destroyed." (Golden 2004, 51). Reasons for the destruction/abandonment of these cities and the associated societal collapse have been attributed to:

**Northern Invasion** The cities of the south were invaded and conquered by nomadic Amorite tribes from the steppes of Syria (Wright 1938, Kenyon 1956). Ben-Tor (1992b, 123)<sup>2.7</sup> observes that, to explain the destruction of some cities and abandonment of others, supporters of this theory propose that cities which were not abandoned 'in terror' were destroyed.

**Egyptian raids** at the end of the Old Kingdom supported by textual evidence from Egypt (Golden 2004, 52) which could have been moti-

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<sup>2.7</sup>Ben-Tor (1992b, 124) suggests that given the lack of strong supporting evidence for any of the theories proposed for the EB IV collapse, an integration of all present (and future) theories should be used.

vated by a desire to discourage Asiatics from moving into Egypt (Mazar 1990).

**Environmental factors** Specifically that there was a general shift in climate to a more arid environment which resulted in a lowering of water tables and significant reductions in the production of food staples causing an exodus from the cities as people turned to nomadic alternatives to feed themselves (Golden 2004, 52). Dever (1989) suggests that there are multiple ecological factors in addition to drought that affected socio-economic organisation, but generally, during periods of extreme drought, these people who turned to nomadism for the procurement of food would have had to compete for resources like water sources and grazing lands, with a highly increased likelihood of conflict which would have in course spilled over back into the settled areas (Golden 2004, 52).

**Human responses** As suggested by Rosen (1995, 27-28) was the inability of the EB III culture to successfully react to a prolonged and general shift in changing environmental factors. This, she suggests, could have been a result of an unwillingness to change their customary response to these conditions, as in the past, fluctuations in weather had set a precedence for the success of particular reactionary steps. Typical reactions, she notes, likely included religious/ritual responses (Rosen 1995, 40). By the time that it was recognised that these traditional responses (which may have been proven successful for many generations) were inadequate, it was too late, and ended with the total collapse of the EB III society (Rosen 1995, 40).

It was not only in the southern Levant where societal disruption was experienced; for Mesopotamia and Egypt (but seemingly not Syria) this was also a period which witnessed unrest and population movements which threatened borders (Ben-Tor 1992b, 124). Given that this trend of collapse and unrest is seen to span regions this large suggests that there was a large, universal event, like a shift in the general climate pattern, underlying these events.

## 2.8 Urbanism

“Urbanism ... A term used by Louis Wirth to denote distinctive characteristics of urban social life, such as its impersonality, but more recently expanded to encompass physical features of urban existence such as can be recovered archaeologically, such as planning, the large size of settlement,

and the presence of religious and political foci alongside residential areas.”

– Darvill (2008, 481)

Broadly described, urbanism can be seen as a focus on cities or *urban areas*, including associated economies, social characteristics, politics, and of course their geography. In addition, consideration is given to the effects on, and caused by, the built environment<sup>2.8</sup>. More specifically to archaeology, the term *urbanism* is used to describe the process whereby populations are caused to live in cities. Cities are often broadly described as having a population of 5000 or more in addition to exhibiting certain traits, such as central administration and the segregation of citizens by some means (e.g. class or occupation). On the topic of urbanism, assumptions abound. It is *easy* to assume, for example, that inhabitants of the earliest cities congregated there for reasons such as protection, or to collectively enable the construction of large, otherwise unachievable projects like irrigation systems or large public buildings; it is **not** easy, however, to find evidence to support assumptions like this. There are basic physical challenges that must be addressed when large numbers of people live together, however, including the production and transport of food/goods, as well as the transport/disposal of rubbish and sewage.

**Table 2.3:** List of Urbanism Models and differentiating features Chesson (2003).

| Models                           |                               |
|----------------------------------|-------------------------------|
| City/Secondary State             | Economic Elites               |
| Dynamic Corporate Villages       | House Society                 |
| Features                         |                               |
| System Of Civic Governance       | Nature Of Residential Systems |
| Nature Of Administrative Systems | Administrative Personnel      |

The topic of urbanism is clearly a very complex issue with many different facets one could focus on. This section presents the view on south Levantine urbanism current until the 1990s, and still adhered to by some scholars today, along with a discussion of Philip and Chessons critique of this traditional view followed by a discussion of the alternative that they propose. The land-use implications of these models have never really been tested, and it will be demonstrated how the Landuse Analyst modelling software developed for this thesis can provide such a test. Chesson (2003, 80) identifies four different models of urbanism according to four main features (see Table 2.3).

<sup>2.8</sup>In this context, I intend for *built environment* to refer to what might be called ‘man-made’ surroundings that provide the setting for human activity. These surroundings (and/or features) can vary in scale from an individual’s own personal shelter to entire villages to monumental public architecture. See Philip (2003, 109-115) for a thorough discussion of settlements and monuments as they relate to the *built environment*.

Chesson provides a succinct description for each (Chesson 2003, Table 1, 80) included here as Table 2.4.

**Table 2.4:** Models of EBA civic governance and nature of social differentiation (after Chesson (2003, 80))

|                                         | City/Secondary State                                                                                                                                                          | Economic Elites                                                                                                                                                                    | Dynamic Corporate Villages                                                                                                                                                                                                          | House Society                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>System Of Civic Governance</b>       | Each community ruled by a king and ruling family, but temple institutions provide a structure complementary to that kingship                                                  | Elite controlled ritual and economic structures, facilitating exchange between urban consumers/traders and rural producers; power shared amongst elites in councils or assemblies  | Power-sharing amongst various groups (e.g., lineages, economic factions, ritual factions) in a heterarchical village community                                                                                                      | Heterarchical power-sharing amongst a community's houses; members of governing body drawn from a variety of houses, thereby cross-cutting lineages, economic groups and other factions within a community                                                                                                                                                                                                             |
| <b>Nature Of Residential Systems</b>    | Palaces housed kings and royal family; residential compounds housed non-elites                                                                                                | Residential compounds housed population, may have been organised into neighbourhoods; elites lived in large residential compounds (formerly interpreted as palaces)                | Residential compounds housed populations; residences functioned as the primary locus of craft production and economic, social and political action                                                                                  | Residential compounds housed populations; households include extended family groups based on real and fictive kinship, and functioned as fundamental unit of social, economic and political organisation; house membership includes many physical households, and may be reflected generally in neighbourhoods within each community                                                                                  |
| <b>Nature Of Administrative Systems</b> | King and ruling family controlled storage and utilisation of administrative complexes; priests of the temple institution controlled the temple's resources and labour sources | Elites controlled craft production (metallurgy, ceramics), agricultural production, exchange of commodities, and ritual institutions                                               | Drawing on resources of cooperative and competitive factions (kin groups, most likely), community leaders deploy collective wealth and assets toward communal projects (fortifications, irrigation, administrative complexes, etc.) | Drawing on resources of cooperative and competitive houses, power-sharing council deploys collective wealth and assets toward communal projects (fortifications, irrigation, administrative complexes, etc.); administrative complexes focused on storage of goods, control of resources (cisterns, granaries) and civic governance; ritual compounds provide areas for ritual practice of specialists and laypersons |
| <b>Administrative Personnel</b>         | Class of managers and bureaucrats to work for the palace or temple institutions to control and redistribute goods                                                             | Elites worked as economic 'middlemen', responsible for organising the creation and maintenance of infrastructure of urban life, craft specialisation, exchange and ritual practice | Community leaders drawn from several kin groups or factions, cross cutting the population                                                                                                                                           | Individuals drawn from houses, cross-cutting lineages, economic groups and other factions in the community                                                                                                                                                                                                                                                                                                            |
| <b>Sources</b>                          | Ben-Tor (1992a), Esse (1989)                                                                                                                                                  | Joffe (1993)                                                                                                                                                                       | Schwartz (1994), Philip (2001)                                                                                                                                                                                                      | Chesson (1997, 1999)                                                                                                                                                                                                                                                                                                                                                                                                  |

### 2.8.1 Conventional Views of Urbanism

“At the beginning of the Early Bronze Age, a relatively low level of political organization and integration most likely existed, but by the [EB III] vast city-states with developed urban centres appeared around the countryside.”

– Golden (2004, 49)

Conventional views maintain that Levantine urbanism<sup>2.9</sup> both followed and echoed Syria and Mesopotamia, but on a smaller scale. Falconer and Savage (1995, 48) summarise the conventional views of urbanism in the southern Levant as “... [being a] secondary, derivative expression of early urbanism.”. They support this summary with three general observations, suggesting that urbanism in the southern Levant has conventionally been:

**A Region-wide Phenomenon** With few exceptions (e.g. Joffe (1991), Finkelstein and Gophna (1993), Gophna and Portugali (1988)), most authors have approached urbanism with a regional view, with ‘consideration for the multitude of “...little discussion of its variable manifestations in the southern Levant’s constituent settlement zones.” (Falconer and Savage 1995, 48).

**An Equivalent to Earlier Models** Although it occurred later and at a smaller scale, urbanism in the southern Levant “... is construed as structurally equivalent to that found elsewhere ...” (Falconer and Savage 1995, 48).

**Associated with Antecedent Forms of Material Culture** Egypt and Syria provide notable examples of regions where material culture and technologies appear similar to those of the southern Levant (e.g. Ilan and Sebbane (1989) and Kempinski (1989)), leading to the assumption that, “Therefore, this material evidence must have spread into the southern Levant, presumably from earlier, exotic sources.” (Falconer and Savage 1995, 48).

Conventional views also talk of settlement fortifications, and larger settlements with longer lifespans, as pointed out by Broshi and Gophna (1984, 41). Broshi and Gophna (1984, 49) propose that these large, fortified centres were possibly places of refuge for smaller satellite settlements in times of emergency, and that the size and location of settlements was dictated by security considerations.

“... At the beginning of the third millennium the population of Palestine was undergoing urbanization. This is not to say that the entire population moved into towns; on the contrary, it may be assumed that the majority continued to live in the tens and hundreds of villages scattered over the

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<sup>2.9</sup>Widely agreed as having occurred in/by the EBA II/III - see Golden (2004, 49) quote, this page.

countryside, as they did in later periods. However, the crystallization of the city or town as a form of settlement is the single most important social phenomenon during this period in the Near East as a whole and appears to be one of the essential conditions for the development of civilization.”

– Ben-Tor (1992a, 85)

Ben-Tor (1992a) attributes the following characteristics as essential for defining a settlement as urban (although he does point out that not all are required in every case): size, fortification, population density, public buildings, social stratification and non-rural pursuits. The city-state model<sup>2.10</sup> is seen as the urban unit which dictated the political organisation of the southern Levant (Ben-Tor 1992a, 86).

Broshi and Gophna (1984, 50) attribute population expansion/intensification in harsher regions (such as the hill country) to the increase in population and a growing need for land. This sentiment is seemingly not shared by Ben-Tor (1992a, 85), who describes settlement *densities* increasing to accommodate the growing population, the growth of which he claims was “...spurred by the improvement in the carrying capacity of the land, that is, by the greater number of persons who could subsist on a given area.” (Ben-Tor 1992a, 85). Ben-Tor fails to support this statement in any way, however. This is important to note, because it is here that the methodology presented in this thesis can be utilised to test what really was just an assumption.

There have been attempts to provide quantifiable arguments for this model. In his 1994 paper, Wilkinson argues that limits of productivity, balanced against demography, favours the formation of hierarchies and ‘caps’ the maximum sizes of settlements. This is a classic explanation of distance decay, where combining time, yields, amount of land, and the number of people *necessitates* a kind of urbanism, namely a classic mesopotamian model. The argument is that a certain population level combined with limited land productivity could *necessitate* an urban system with some sites producing surplus food to another site that is unable to *support* itself, but does provide secondary products to the supporting settlements, facilitating in their ability to produce this surplus. Wilkinson presents a model showing how EBA site and population sizes reflect ecological and economic constraints of their hinterlands by assessing the productivity of the lands surrounding the settlements. The productivity and availability of arable land is applied in Wilkinson’s model as a constraint, when, in fact, it is equally possible that it could have been the opposite, and encouraged population growth. Wilkinson (1994, 484) admits that it was necessary for him to restrict certain aspects of the data used in his model to address the broad range of issues in the paper. Sixteen years on, can the more powerful tools available to us enable these restrictions to be lessened? The methodology and software written for this thesis attempts to directly address this

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<sup>2.10</sup>where a town or city is surrounded by smaller satellite settlements which formed its agricultural hinterland and for which the central settlement provided protection and support

problem Wilkinson faced in 1994.

By carefully modelling the subsistence strategies, diets and land-use of communities on a regional scale, it will be possible to see if an increased carrying capacity was required or even necessary. If the area of land required to sustain a number of contemporary adjacent settlements was more than was available (given calculated population levels), then there would be an argument for an increased carrying capacity; conversely, it has been proposed that an increased population must have been supplemented by nomads and semi-nomadic people, although “... it may be assumed that [the number of people] did not exceed some thousands or a few tens of thousands.” (Ben-Tor 1992a, 85).

By examining the dietary needs for the settlements in a region, and calculating their land requirements for producing enough food to sustain themselves, it becomes possible to estimate what, if any dependency these large settlements may have had on putative ‘nomadic’ people. These ideas are discussed in more detail in Chapter 6, where regional network analysis is carried out.

### 2.8.2 The Corporate Village

For those who have seen the Early Bronze Age as an essentially urban period, some have seen the EB I as the first phase in the emergence of urbanism (Kempinski 1992) and yet others as essentially an uncomplex village society (Braun et al. 2001). In order to gain further insight into the developments leading up to what some call urbanism (which is more traditionally associated with the EB II-II in the Southern Levant) there has been a focus in this thesis on the transformative EB I period. A better understanding of this transformative period will also help to provide a basis upon which a better evaluation of what might be called the Falconer-Philip-Chesson (FPC) model of non-urbanism in the Early Bronze Age of the Southern Levant.

Philip and Chesson suggest a heterarchical<sup>2.11</sup> corporate village model or house society (Chesson 2003) for much of the EBA and EB I probably represents this in their view.

At the same time Philip argues for innovation within such societies and sees EB I as an example of this, particularly in relation to relationships with the landscape. He sees fortified settlements as part of the way in which large communities<sup>2.12</sup> inscribe themselves on the landscape and these start in EB I (e.g. Jawa and possibly Pella).

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<sup>2.11</sup>In a ‘social’ context, a *heterarchy* distributes privilege and decision-making among members of society. A *hierarchy* on the other hand allocates power and privilege to members perceived as being higher in the structure. In a ‘systemic’ perspective, Probst et al. (1992, 127-132) describe heterarchy as being a more flexible version of otherwise formal relationships inside an organisation whereby links can be reversed or privileges redistributed (e.g. domination/subordination) in given situations following the needs of the system.

<sup>2.12</sup>large sites certainly appear; Shuna grows to 20 hectares, as does Beth Yerah/Khirbet Kherak, Erani is probably 17 hectares, and Megiddo between 20 and 40 hectares.



Philip sees an intensification and integration of aspects of the agricultural economy. He suggests more widespread adoption of irrigation (McCreery 2003) plough agriculture, fruit tree cultivation (including the appearance of cultivated vines in Southern Levant), and large scale employment of donkeys as pack animals (in particular, numbers are high at Afridar and Fyanan for example). Water management systems are suggested for Jawa, although these are not easy to date. Evidence for plough cattle is apparent in figurines and cattle pathologies as well as age profiles this may be some of the best evidence for an intensification. Does this imply more terraces, built irrigation systems he certainly seems to think so given his concept of communities inscribing themselves on the landscape but is there evidence of this, or of the scale of their impacts.

In discussing the range of social, political and economic structures of EBA life, Chesson (2003, 79) proposed what she referred to as an intriguing theoretical puzzle in the form of two questions: "... is it possible to encompass the variation of EBA 'urban' lifeways, social networks and political structures within a single explanatory model? Can any reconstruction accommodate the range of cultural expression that we define as EBA society?" The methodology developed for this thesis allows for a more quantifiable approach to this 'theoretical puzzle', and it is evident throughout Chapters 5 and 6 how these questions can begin to be addressed by systematically modelling a range of landuse and subsistence scenarios.

There has been work done that can be interpreted as to providing some quantifiable support for this explanation of social development. Rank size analysis uses a regional approach to attempt to determine if the location of settlements vary from a predicted pattern of rank with respect to size (Darvill 2008, 377). Essentially, if the result is linear, it suggests that the rank-size relationship indicates a central place system, interpreted as urban. A convex curve, on the other hand, occurs when medium sized sites are larger than expected, suggesting a series of independent medium sized settlement, thus indicating a non-integrated (and thus) non-urban system. When applied to the Southern Levant, this type of analysis (Harrison and Savage 2003, Savage and Falconer 2003, e.g.) seems to provide support for the 'Corporate Village'. It is important to note, however, this approach is quite coarse, not incorporating food requirements, land availability, population, or other potentially important factors. Further, it assumes a region-wide market system, thus failing to address the possibility of localised urban clusters. Clearly, a more robust model is possible by incorporating these factors; indeed, the methodology<sup>2.13</sup> presented in Chapter 4 of this thesis *does* incorporate population, food requirements, land availability, population levels and more into a regional study (see Chapter 6).

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<sup>2.13</sup>together with the Landuse Analyst software that has been written.

## 2.9 Conclusion

Fundamentally, economic organisation can be thought of as the strategy used to collect and distribute resources. As Clark (1976) points out, however, a society's economy is much more than this; people are, after all, quite irrational. Taboos and religious restraints, for example, might prevent an abundant resource from being used, or perhaps some resources are just favoured in general. Clark (1976) offers two approaches regarding the study of economic strategies. At the most basic level, it is suggested that the approach be based upon time/labour division, role allocation, resource distribution, and the control/regulation of production. The second approach is a study of subsistence methods (Clarke 1968).

A close examination of subsistence strategies and associated engagement with the landscape has the potential to provide additional clues for the cause of the collapse of EBA societies. If there was a shift in climate to a period of greater aridity, certain subsistence strategies will be better able to handle this change, whilst others could be potentially disastrous. For example, if society had turned their focus away from meat and were heavily dependent on cereals, herd sizes would have become substantially reduced, increasing the vulnerability of the food supply to drought. An increased reliance on dairy and cereals and reduced meat would mean large herds, but foddering would have been likely, as in order to get the milk, it would have been preferential to have the animals closer to the settlements; as will be shown in Chapter 5, foddering, whilst greatly reducing the amount of grazing land required, results in an increased amount of crop land (and reliance upon cereals) again making the food supply vulnerable to sustained drought. This also greatly increases the likelihood of conflict in widespread drought conditions, because if the herds' grazing range had to be extended to keep them fed (and sustain the production which communities had come to rely upon) competition for this additional land could have been intense.

By incorporating as much archaeological evidence as possible into a detailed examination of potential subsistence strategies, it is possible to show that certain strategies more closely fit (or do not fit) the events which are known to have occurred. Incorporating all of this evidence into reconstructing potential scenarios, however, results in an extremely complicated, intertwined model that can result in an almost infinite number of different scenarios. As a result of trying to develop a methodology that incorporates as much of the data available from the archaeological record as possible for use in this research, it became evident that it would require a fast and systematic tool in order to examine the countless number of scenarios which could be derived from the input data. In response to this, Landuse Analyst was developed. This sophisticated land use modelling software has been specifically designed for use across all regions and time periods, and is one of the primary contributions of this research.

This chapter has explained in brief the Chalcolithic and Early Bronze Age of the southern Levant and illustrated why a more nuanced understanding of the important transformative EBA I in particular will allow for a better evaluation of the various models of urbanism proposed for the Southern Levant. In the following chapter, a brief history of modelling, site catchment analysis, and Geographical Information Systems will be presented to set the stage for Chapter 4, where the development and implementation of Landuse Analyst is presented.



## Chapter 3

# A Brief History of Site Catchment Analysis

“It therefore seems (regrettably) necessary to coin an independent name for the study of the relationships between technology, and those natural resources lying within economic range of individual sites. We propose the term ‘site catchment analysis’.”

– Vita-Finzi and Higgs 1970, 5

### 3.1 Introduction

Chapters 1 and 2 introduced the topic of research being examined in this thesis, provided a brief description of the southern Levant, and outlined some of the more pertinent archaeological background information. Before presenting the methodology used in this research, a brief overview of the development (and methods) of site catchment analysis will be given in this chapter. The purpose of this overview is to provide the reader with a background with which they can compare the methodologies developed and presented in this research with those undertaken by previous researchers. Coupled with the discussion of site catchment analysis will be a discussion of Geographic Information Systems and predictive modelling, and the roles which they have played in the history of archaeological analysis.

Once this background information has been presented, the methodology will be presented in detail in Chapter 4 (page 51). As well, it provides a context for a further extension of this methodology presented in Chapter 6 (page 325). This shows how catchment analysis can be effectively integrated into an analysis of networks on a regional level.

Innovations in technology and its spread in society can be divided into four phases (Foote and Lynch 2000):

**Initiation** Innovation makes something available for the first time.

**Contagion** Experimentation follows exploring how the innovation can be changed to accommodate the needs of research/commerce (it is not necessary for all of these experiments to work, but at least some must.)

**Co-ordination** The best applications of the innovation gradually gain acceptance and are developed collaboratively; experiment co-ordination results in a distribution of any costs of development and implementation.

**Integration** The innovation gains widespread acceptance and is thus integrated into mainstream research.

Technology has played an important role in advancements of archaeological investigations. In the following discussion, it can be seen how the technological advancements relating to site catchment analysis can, for the most part, be seen to have progressed through the sequence outlined above.

### 3.2 Origins of Site Catchment Analysis

The concept of site catchment analysis was first proposed by Vita-Finzi and Higgs (1970) as the “...*study of the relationships between technology and those natural resources lying within economic range of individual sites.*”. Their process of site catchment analysis, as first presented, has been summarised by Bailey and Davidson (1983, 89) as:

1. defining a problem in the archaeological literature at the time, with a proposed methodological solution.
2. drawing upon ethnographic work and fundamental energetics (e.g. Chisholm 1968 and Lee 1969), stating as fact that there are always some spatial limitations to the field of potential resources to which the inhabitants of a given community had access.
3. identifying several resource categories (i.e. arable land, grazing land) and assigning to them variables relevant to the investigation of site exploitation territories.

The very basis of site catchment analysis relies on the assumption that “... the further the area is from the site, the less it is likely to be exploited ...” (Vita-Finzi and Higgs 1970, 7)

In nearly all academic papers regarding site catchment analysis, two references are consistently used to support the above assumption: *!Kung Bushman Subsistence: An Input-Output Analysis* by Lee (1969) and *Rural Settlement and Land Use* by Chisholm (1968). Lee, in an ethnographic study of the Kalahari Bushman hunter-gatherers, finds that women do not generally forage for food (plants) beyond 10 km from their main camp, and although the men sometimes travel further when hunting, they generally

set up a temporary over-night camp when they do. Chisholm, on the other hand, finds that there is a general 3-4 km limit between farmers and their fields (Chisholm 1968, 62;125;146;147), noting that only exceptionally do farmers travel further, and that “at a distance of 3-4 kilometres the costs of cultivation necessitate a *radical* modification of the system of cultivation or settlement – for example by the establishment of subsidiary settlements – though adjustments are apparent before this point is reached.” (Chisholm 1968, 148). It was studies like these that prompted Vita-Finzi and Higgs to use what seems to have become the now de-facto standard catchment radius of 5 km. The reasoning behind this was that the resources the community would have required would have come from the 7854 ha ( $\pi \cdot 5^2 = 78.5398163 \text{ km}^2$ ) catchment that the 5 km limit encompassed which the authors took as “... the economic threshold for the more intensive subsistence agricultural economies.” (Vita-Finzi and Higgs 1970, 16). By examining the land in the now ascribed catchment area, they were able to classify the land into the following categories: arable, potentially arable, grazing, marsh and dunes.

Each category was then expressed in terms of percent of the total catchment. By examining the ratio of the weight adjusted land types, they drew conclusions regarding site function (e.g. if a site’s catchment area contained low percentages of arable land the settlement’s subsistence strategy was deemed to be largely non-agricultural) (Flannery 1976a, 92). It is important, however, to draw attention to the fact that 20% of 7854 ha is 1570.8 ha; problematic because 1570 ha was more land than was required, a fact which was not lost to critics of their original methodology.

The original methodology for site catchment analysis by Vita-Finzi and Higgs (1970) included a rudimentary method for dealing with travel time. Vita-Finzi and Higgs recognised that walking time was an important factor for determining the site catchment boundary. They did not directly apply this to agri-

| Distance in kilometres | Net Output (after Virri 1946, 7) | Arbitrary Weighting adopted (as percent) |
|------------------------|----------------------------------|------------------------------------------|
| 0-0.1                  | 100                              | –                                        |
| 0.5                    | 67                               | –                                        |
| 1.0                    | 50                               | 100                                      |
| 1.5                    | 40                               | –                                        |
| 2.0                    | 33                               | 50                                       |
| 3.0                    | 25                               | 33                                       |
| 4.0                    | 20                               | 25                                       |
| 5.0                    | 17                               | 20                                       |

**Table 3.1:** Weightings for agricultural catchment bands (after Vita-Finzi and Higgs 1970, Table 3)

cultural catchments but instead drew on work done by Virri (Vita-Finzi and Higgs 1970, 29), who found that, in Finland farmsteads, both net and gross yields per hectare decreased with the distance from the farmstead. To compensate for the exclusion of walking time in their methodology for agricultural land catchments, they instead implemented a system of weighted concentric circles drawn around sites. This weighting factor was applied at increments of 1 km starting at the settlement and stopping when the radius reached 5 km. The categorised land contained within each band formed by

the concentric circles was then given a weighting (effectively this is of diminishing productivity), and the potential contributions from the resource contained in each band was then reduced accordingly. The weighting factors are shown in Table 3.1.

There is a potential problem with the applicability of the work Virri did that, to my knowledge, has not been satisfactorily addressed<sup>3.1</sup>. The issue I have with the progressively diminished agricultural productivity is that Virri (1946) was talking about *farmsteads* and not towns or cities. Of course it makes sense that a farmer will be unable to give as much attention to fields which are further away from his home in comparison to those that are close; applying this to ancient settlements where hundreds and thousands of people lived together in a settlement, however, is in my opinion a completely different thing. The person who is tending their crop will have to walk from their residence in the settlement (where they reside with everyone else) to and from their land, and unless every farmer in the settlement had an incredibly narrow pie shaped wedge of land originating at the settlement and consistently expanding directly outwards, this methodology should not be applied. On the contrary, I would argue that as the distance from the settlement increased, productivity might actually increase. This position is based on the principle that cultivation of land would occur on an as needed bases, therefore the ‘newest’ cultivated land would most likely have been that land further from the settlement, as it would have been in use for less time than the land immediately surrounding the site, thus being more fertile and less depleted. At any rate, until there is evidence of ancient farmers living in the centre of their fields, the methods for catchment analysis proposed in this thesis seem to be a much more suitable approach than that of progressively diminished agricultural productivity based on distance from a settlement.

Site catchment analysis has been continually evolving with various refinements, enhancements and additions since its conception. One important development occurred in 1976 when Flannery (1976a) presented additional variants of site catchment analysis. After initially presenting the San Lorenzo case study (contributed by Rossmann 1976, 95) using an implementation of the original format pioneered by Vita-Finzi and Higgs, Flannery himself presented a section on Oaxaca and Tehuacán where he reversed the process by examining the empirical data of the resources found at the site and determining where they came from. Using this approach, he avoided the problem of defining site function based on landscape, which was one of the often repeated criticisms of the Vita-Finzi and Higgs methodology (for examples see Roper 1979, R.W. Chapman in Clarke 1978, 125-128, Hodder and Orton 1976 and Wobst 1978). This is

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<sup>3.1</sup>The Virri (1946) paper was difficult to get. Only in the final 3 weeks before submitting this thesis was I able to locate it, thanks to the help of Anu Riikonen who was able to finally locate it in the University of Helsinki Library archives. Further thanks to Anu for his help with the translation of the paper. Finally, one reason for the difficulty of finding this paper was due to the fact that previous bibliography entries have been (consistently) misspelled, without a Volume number, and with an incorrect journal name as well.



important and directly relevant to this research because it is this ‘what is here, and how can we use it?’ approach which has largely been taken throughout this thesis.

For further examples of early site catchment analysis methodologies, the reader can be directed to several sources (see Table 3.2).

**Table 3.2:** Early publications focussing on different catchment analysis methodologies.

| <b>Methodological Focus</b>                                                                                    | <b>Reference</b>                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Weighting of sites                                                                                             | Dennel and Webley (1975)<br>Peebles (1978)<br>Webley (1972)                                                                                |
| Off/on site comparison quantiles                                                                               | Akazawa (1980)<br>Bailey (1978)<br>Bailey (1982)<br>Davidson (1976)<br>Hastorf (1980)<br>Jarman et al. (1976)<br>Munson and Yarnell (1971) |
| Predictive modelling of site locations                                                                         | Bailey (1978)<br>Gamble (1979)<br>Jochim (1976)<br>Schwellnus (1974)<br>Sturdy (1975)                                                      |
| Analysis of locations relative to each other<br>checking assumptions about territory size and site<br>function | Brumfiel (1976)<br>Cassels (1972)<br>Dennel and Webley (1975)                                                                              |
| Regional analysis as a check on territorial site data<br>interpretation                                        | Bay-Petersen (1982)<br>Gamble (1978)<br>Sturdy (1975)<br>Zarky (1976)                                                                      |

Bailey and Davidson (1983, 88) define six objectives of site catchment analysis as:

1. defining areas regularly used by the inhabitants of a settlement for their day to day subsistence
2. tracing materials and resources present in the archaeological assemblage to their points of origin in the surrounding landscape
3. reconstruction of micro-environments surrounding a settlement to gain insight into variations of palaeoenvironmental data evident in the archaeological record.
4. reconstructing the potential food resources which would have been available to the inhabitants of the settlement, which infers the subsistence economy they practised.

5. reconstructing the possible function of settlements
6. reconstructing socio-economic ties between settlements when considering them as members of regional settlement systems

It has been important to outline the origins of site catchment analysis as a methodological/procedural tool because it shows how the process was limited by technological shortcomings. These shortcomings have been vastly reduced by developments in both hardware and software, and the following section describes how GIS has become an indispensable tool in site catchment analysis.

### **3.3 Geographic Information Science**

Today's commonly used technique of utilising aerial photography and satellite imagery for archaeological investigation began in 1922 (Olsen 1985). It is now pertinent to examine how remote sensing in archaeology spawned a reactionary process which resulted in the development of one of archaeology's most powerful analytical tools: the modern day Geographical Information System.

#### **3.3.1 Remote Sensing**

Remote sensing made it possible to survey areas much more quickly. Remote sensing also made it possible to conduct surveys in areas where the landscape was difficult or impossible to traverse because it could now be done without being physically on the ground. The resultant images provided an efficient way of storing/recording vast amounts of data for large tracts of land. Countless other advancements in technology also contributed to this burgeoning pool of data from which archaeologists could draw. The problem was, however, that there were no effective tools for working with this vast amount of data which meant that remote sensing could only make a limited contribution to archaeological knowledge; with the advent of computers, archaeologists were given the tool that enabled them to finally make use of the available data.

One tool particularly effective for working with high levels of spatial data in a fast, systematic and efficient manner is the Geographic Information System (henceforth GIS) which allows for a rapid and systematic way to combine spatial data from a wide number of different sources. Combining spatial data was first illustrated in 1950 when Jacqueline Tyrwhitt took data regarding elevation, hydrology, geology, and cropland and combined them into a single map (Tyrwhitt 1950).

#### **3.3.2 Map Overlays**

The pioneering work done by Tyrwhitt was done entirely by hand, literally cutting sections from one map and lining them up on another, thus superimposing the cut-outs

as ‘map layers’. Nine years later, this task was able to be done with a computer thanks to Tobler who developed MIMO (Map In-Map Out). MIMO used a three step process: map input, map manipulation, and map output (Tobler 1959). Although not powerful by today’s standards, this pioneering effort served to simplify the process of map overlays. As map overlays became easier and more common, it triggered further development and resulted in, among other things, the United States Central Intelligence Agency’s *World Data Bank* and the US Census Bureau’s *Dual Independent Map Encoding* (DIME); a revolutionary system for encoding digital maps that is still in use today (Clarke 1997, 8). The rapidly increasing volume of digital data spawned the development of tools to deal with it. One of the first GIS programs developed was the Canada Geographical Information System, or CGIS, introduced in 1964.

### 3.3.3 Defining GIS

Fundamentally, a GIS is a tool which allows for the manipulation of spatial data. However, a GIS provides further analytical possibilities. Because of the highly adaptable nature of a GIS, it is difficult to provide a single explicit definition. An example of a simple definition is:

“A powerful set of tools for storing and retrieving at will, transforming and displaying spatial data from the real world for a particular set of purposes”

– Burrough (1986, 6).

The above definition provides no clues as to what the purpose of a GIS could be, nor does it explain how the data is represented or stored. This definition is contrasted with:

“A geographic information system (GIS) is an information system that is designed to work with data referenced by spatial or geographic coordinates. In other words, a GIS is both a database system with specific capabilities for spatially-referenced data, as well [as] a set of operations for working with data . . . In a sense, a GIS may be thought of as a higher-order map.”

– Star and Estes (1990, 2-3).

Star and Estes provides a more explicit definition including a more precise definition of the type of data a GIS can work with. A more succinct definition is:

“A geographic information system is a special case of information systems where the database consists of observations on spatially distributed features, activities or events, which are definable in space as points, lines, or areas. A geographic information system manipulates data about these points, lines and areas to retrieve data for ad hoc queries and analyses”

– Duecker (1979, 106).

Duecker addresses how data is represented, and expands on the type of data suitable for inclusion in a GIS, but fails to convey the broader potentials of a GIS (e.g. modelling).

An important component of any GIS is its database, but a database in a GIS is different from a 'normal' database: it is 'spatially aware' (capable of determining spatial relationships with data). For a database to be spatially enabled, it must incorporate (Wheatley and Gillings 2002, 23) a:

- **locational component:** the 'position' in terms of geographical space and determines where (and in what form) things are.
- **topological component:** referring to the logical relationships between the various objects in the geographic database.
- **attribute component:** those features (or characteristics) of objects that determine what things represent and their properties.
- **metadata component:** the detailed documentation providing all relevant information pertaining to each object in the spatially enabled database

A GIS can combine different types of spatially-referenced data that can result in new outcomes, meaning that a GIS can also be described as a tool capable of producing new information. The new information provided by GIS can contribute knowledge that was previously unobtainable.

### 3.3.4 GIS in Archaeology

The use of the term 'GIS' began appearing in archaeological publications between 1983 and 1985 (Kvamme 1996). As the focus in archaeology began shifting from the settlement to the landscape (Warren 1990a), and data was becoming more readily available for immediate analysis, there was a growing need for automated predictive modelling. Predictive modelling establishes a 'causal' relationship between archaeological site locations and various environmental factors upon which a statistical model is based (Gourad 1999). Computers became the predominant tool for creating the statistical models that become a convention in archaeology (Aldenderfer 1987). GIS is a very effective tool to use for predictive modelling, and it is perhaps this feature of a GIS which more than anything made GIS a mainstream tool in archaeology (Gourad 1999). With the introduction of relational databases, archaeology as a discipline was impacted by changing the way that data was collected (e.g. remote sensing), the nature of questions regarding the data (e.g. the ability to explore complex logistical regression models), and, as some have suggested, even the general conceptual design of an archaeological dig (Harris and Lock 1996, 352).

### 3.4 Modelling

Sebastian and Judge (1988) suggested that predictive modelling became increasingly important as archaeology went from a largely descriptive stage to a more explanatory stage.

Drawing on the mathematical modelling undertaken by Orton (1980, 20), Voorrips (1987, 61) defines archaeological models as “... [the] partial representations of a theory ... formulated in a manner which enables the archaeologist to test the theory by means of empirical data”. Warren describes predictions as a critical component of scientific method; hypotheses are formed to explain new patterns found in various contexts, and models are then built in different contexts in order to test the hypothesis (Warren 1990b).

Regression analysis is a statistical technique for determining the relationships that exist between variables. There are several kinds of regression analysis (e.g. linear, non-linear, stepwise) and predictive modelling in archaeology draws upon all of them. When using a two variable method (bivariate regression<sup>3.2</sup>), one variable is known as an ‘independent’ variable (e.g. proximity to a water source) and the other variable is a ‘dependent’ variable (e.g. number of sites). A value is *chosen* for the independent variable, and the value for the dependent variable is then calculated; for example, if determining how many sites (*s*) are located within *x* km of spring, the independent variable is the distance from (or proximity to) a spring (*x*), and the dependent variable is the number of sites found (*s*). When the complexity of a model increases, and more variables are introduced, it is called multivariate regression. Multivariate regression models look at the effects of one independent variable on multiple dependant variables. The next level of regression modelling becomes much more complex; logistical regression models examine the relationships between **multiple independent** variables and **multiple dependent** variables. To make this even more complex, each independent variable can have as many parameters as the archaeologist wants (Savage 1990). *Step-wise* logistical regression is the extension of logistical regression, different only because it uses pre-selected significance levels where variables get added or removed (Christopher et al. 1996). The methods best suited to archaeological applications are the two latter (and most complicated) methods and also describes the type of modelling used throughout this thesis. The above discussion describe a few select examples highlighting some of the early applications of computers in archaeology.

The examples discussed above are drawn from published work that is replete with diverse examples that utilise the continually increasing power of both computer hardware and software (e.g. Bevan 2008, Fletcher 2008, Kvamme 2008, Fitzjohn 2007, Wilkinson et al. 2007, Kvamme 2007, Given et al. 2007, Ebert 2006, Hare 2004, Baja

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<sup>3.2</sup>Regression is defined as “a measure of the relation between the mean value of one variable (e.g. output) and corresponding values of other variables (e.g. time and cost).” – Jewell et al. (2001)

et al. 2001, Ebert 2000).

### **3.4.1 Modelling Landuse in GIS**

In the last 7 years, there have been a number of interesting attempts to model past land use and landscape evolution in a GIS environment, and when broadly classed can be placed into one of four approaches; namely geomorphological modelling, agent-based modelling, predictive/locational modelling and site catchment modelling. This section (3.4.1) presents and briefly discusses one example for each of these four approaches.

#### **Geomorphological Modelling**

Ayala and French examined landscape development of north-central Sicily, integrating a geomorphological model<sup>3.3</sup> of the Fiume di Sotto di Troina valley with archaeological survey in a GIS to model past landuse practices of the Roman and Medieval periods and the possible potential erosion caused by different land use practices throughout history (Ayala and French 2005). An important outcome from this work was that their erosion models showed “[how intensive geoarchaeological survey]... can aid in the interpretation of occupational history in a way that neither geomorphological nor archaeological investigation can accomplish on their own.” (Ayala and French 2005, 165). The ability of Landuse Analyst (see Section 4.3 on page 80) to more accurately and systematically model a settlement’s specific catchment area and land use strategies would have been a valuable tool for inclusion in their methodology, and in fact is an interesting potential future collaborative project which could be explored.

#### **Agent Based Modelling**

In his 2008 paper, Wainwright (a geographer at the University of Sheffield) uses an agent based model<sup>3.4</sup> to help understand human-landscape interactions. Basic actions like the acquisition of food are simulated to the scale of the individual. The CybErosion modelling framework he presents attempts to include human actions into a landform evolution model by simulating how people interacted with their landscape (Wainwright 2008, 659). It would be very interesting to incorporate this level of human modelling into the methodology and software developed for the research presented in this thesis, and in doing so, give the agent based model a further platform by which it could include many more features provided by Landuse Analyst, such as crop/herd/dairy modelling and regional network analysis.

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<sup>3.3</sup>Geomorphological models focus on changes in the physical (geomorphological) landscape.

<sup>3.4</sup>Agent-based modelling focuses on building large-scale models from simulations of the behaviour of many individual agents.

## **Predictive/Locational Modelling**

Legg and Taylor examined the influence of environmental factors on the distribution of Irish ringforts in the Inny River catchment area. Along with environmental variables, locational information was entered into a GIS, and a model was created for predicting the density of ringforts. When extended outside of the Inny River catchment area, their model succeeded in producing results that were considered broadly consistent with recorded site distributions (Legg and Taylor 2006, 201, 214–215). The methodology developed in this thesis for both the network and site catchment analysis would provide an interesting way to look at how the distribution of these ringforts could also have been linked to subsistence strategies, and in combination with their work, possibly result in a more robust and accurate model<sup>3.5</sup>.

## **Site Catchment Modelling**

Perhaps the most pertinent piece of research for my research in the last 7 years, however, has been carried out on Neolithic Calabria in southern Italy by Van Hove (e.g. Van Hove 2002, 2003a,b, Robb and Van Hove 2003).

Robb and Van Hove (2003) explored the relationships between Neolithic land use and culture in southern Italy. In this paper, they focus on how cultural choices affected land use and the resulting human landscape. GIS was used to "...trace out the implications of this cultural choice systematically." (Robb and Van Hove 2003, 242). They attempted to reconstruct land use practices and what the resulting landscape would have looked like by identifying the specific areas which would have been utilised for different agricultural activities (e.g. herding/foraging, farming). They did this in an effort to provide support for contrasting theories of the Neolithic economy and culture.

Similar to the approach taken in this thesis, reasonable estimates of many input factors were used, and the effects of varying these values observed in lieu of trying to determine one single 'perfect' model (Robb and Van Hove 2003, 245). Land classification was done using zones derived from topographical maps, and further classified with a heavy reliance on slope (Robb and Van Hove 2003, Figure 2, page 246). What this study failed to do, however, was to provide a quantifiable method, based on archaeological evidence, to calculate food production and land requirements. Instead, land area requirements were based upon estimations of Neolithic agriculture: "The next step was to estimate the resources a group of 50 people would have required annually to survive. A family of 4-6 people living primarily on grain would have required about 1000kg per year ... and Neolithic grain yields are thought to have been between 500 and 1000 kg/ha ... [so] a group of 50 would have required perhaps 10-15 ha of active grain plots with an additional amount in fallow fields and gardens of other crops."

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<sup>3.5</sup>This potential is pointed out in the conclusions of this thesis in Chapter 7, page 360

(Robb and Van Hove 2003, 246).

In order to locate the specific areas of probable farming land a simple slope based cost surface ( $friction = \log(slope + 1) + .5$ ) was used to calculate travel effort. Further, 20 permutations of the model were examined (Robb and Van Hove 2003, 247). They were able to conclude that “The most important point of our reconstruction of Neolithic land use is that arable land was probably never the limiting resource on Neolithic settlement...”(Robb and Van Hove 2003, 247).

Significantly, these conclusions were based on unquantifiable estimates<sup>3,6</sup> of many of their input factors. In addition, their model is limited by the fact that a static number of 50 was used to model population, and that their 20 permutations were effectively based on three input variables (percentage levels of crops, herds and wild-resources). More significantly, the 20 permutations were determined using values which they felt fit the description assigned to the permutation. For example (Robb and Van Hove 2003, Table 1, page 248) labels the 20 Models (permutations) as Foragers, Forager-Farmer, Farmer, Pastoralist and one as Mixed. A more comprehensive model, like that developed in this thesis, would have proved a much more meaningful and robust method to calculate the total land needs. It would also avoid the problem of assigning input values based on what we feel suit a certain strategy; rather, the model outcome could be analysed, and the labelling could be done based on this. In other words the strategies would be defined by the data, not by the pre-conceived notion of what certain strategies are.

The methodology and software developed for and implemented in this thesis attempts to directly address the problems that using unquantifiable input variables presents to our understanding of past land use. Specifically, algorithms have been developed for the calculation of many of the input variables which in previous studies were simply estimated. The remainder of this thesis is dedicated to the presentation and implementation of this methodology.

### 3.5 Conclusions

In this chapter, the origins of site catchment analysis and the basics of computer modelling in archaeology have been discussed. Site catchment analysis, as first proposed for use in site catchment analysis by Vita-Finzi and Higgs (1970), has been employed with few modifications by researchers right up to the present day (e.g. Campbell 2009, 111). The reader will see, in the following chapter where the methodology of this research is presented, how the use of GIS and logistical regression analysis modelling have been combined into a versatile tool which provides a fast, systematic and improved approach to modelling subsistence catchment areas. The effectiveness of this

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<sup>3,6</sup>I wish only to point out that these estimates are unquantifiable; they were all well referenced



tool will be evident in Chapters 6 and 7, where it is employed to analyse Chalcolithic and EBA Tel esh Shuna North and its surrounding region.

